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**An Analysis of Player-Formation
Graphs for Predicting Football Passes**

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**Analiza grafov postavitv igralcev za
napovedovanje podaj na nogometnih
tekmah**

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Vid Stropnik, 2024

Babici.

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List of used acronyms

acronym	meaning
SNA	social network analysis
GNN	graph neural network
DL	deep learning
GAT	graph attention network
GCN	graph convolutional network
MLP	multi-layer perceptron
ReLU	rectified linear unit
NF(s)	node feature(s)
EF(s)	edge feature(s)
Opps	opponents
PC	pitch control
FIFA	International Federation of Association Football
UEFA	Union of European Football Associations
KDE	kernel density estimation
PV(s)	player vector(s)
PDF	probability density function
SGD	stochastic gradient descent
(L/C/R)(W)B	(Left / Central / Right) (Wing) Defender
(L/C/R)(D/A)M	(Left / Central / Right) (Defensive / Advanced) Midfielder
(L/R)W	(Left / Right) Winger
(L/C/R)F	(Left / Central / Right) Forward
GK	Goalkeeper

Abstract

Title: An Analysis of Player-Formation Graphs for Predicting Football Passes

In modern association football, data and the knowledge derived from it play a crucial role in forming tactical plans and analyzing games. In this Master's thesis, we explore the capability of modeling a passer's decision-making when selecting a target during a football match, using graphs and graph neural networks. We present a methodology for constructing a dataset of graphs from open data, where nodes represent players on the field and edges represent interactions between them. We investigate three different graph configurations and analyze the utility of advanced features, such as player movement trajectories and pitch control models.

Our final dataset contains 80,332 graph representations from 166 matches. We evaluate the usefulness of the dataset on three tasks: regression of the coordinates of a successful pass, classification of the role of the pass recipient, and prediction of the target zone of the pass, based on the positional play pitch division. Our models, based on the Graph Transformer architecture, achieve the best results in all tasks.

The results demonstrate that our approach effectively models the passer's decision-making and that graph models enriched with appropriate features can significantly contribute to advanced football analytics.

Keywords

association football, soccer, graph neural network, graph dataset

Povzetek

Naslov: Analiza grafov postavitve igralcev za napovedovanje podaj na nogometnih tekmah

V modernem nogometu imajo podatki in iz njih pridobljeno znanje ključno vlogo pri oblikovanju taktičnih načrtov in analizi iger. V tej magistrski nalogi raziskujemo zmožnost modeliranja podajalčevega odločanja pri izbiri tarče med nogometno tekmo z uporabo grafov in grafovskih nevronske mreže. Opišemo metodologijo za konstruiranje podatkovne zbirke grafov iz odprtih podatkov, v kateri vozlišča predstavljajo igralce na igrišču, povezave pa interakcije med njimi. V delu med seboj primerjamo tri različne grafovske konfiguracije in analiziramo uporabnost naprednih značilk, kot so vektorji igralčevih premikov ter domenski modeli nadzora igrišča.

Naša končna podatkovna zbirka vsebuje 80,332 grafovskih ponazoritev, vzorčenih na 166 tekmah. Uporabnost podatkovne zbirke evalviramo na treh nalogah: regresiji koordinat uspešne podaje, klasifikaciji vloge prejemnika podaje in napovedovanju ciljnega območja glede na specifično delitev igrišča. Naši modeli, osnovani na arhitekturi grafovskega transformerja, dosegajo najboljše rezultate pri vseh nalogah. Rezultati kažejo, da naš pristop učinkovito modelira podajalčevo odločanje in da lahko grafovski modeli, obogateni z ustreznimi značilkami, pomembno prispevajo k napredni nogometni analitiki.

Ključne besede

nogomet, grafovska nevronska mreža, grafovska podatkovna zbirka

Razširjeni povzetek

Podatki in iz njih pridobljeno znanje v modernem nogometu igrajo veliko vlogo. V vodilnih moštvih jih trenerji in analitiki uporabljajo za pripravo taktičnih načrtov [2], nepogrešljivi pa so tudi v svetovih športnega novinarstva in športnih stav [3,4]. Pri iskanju podatkov za pripravo vpogledov, ki jih vsi omenjeni deležniki tako močno cenijo, se lahko srečamo z različnimi oblikami prikazovanja in povzemanja dogajanja na igrišču. Med njimi so najpogostejši tabelarični podatki, pridobljeni z namenskimi senzorji ali z vzorčenjem video prenosov nogometnih tekem [5–7]. Tovrstne ponazoritve se po nekaterih virih izkažejo za omejujoče – ne samo, da dinamika gibanja nogometašev močno oteži enlično diskretizacijo dogajanja [8], temveč z njimi prav tako ne moremo celovito prikazati medsebojnih vplivov igralcev na soigralce in tekmece [9]. Za veliko primernejše se zato izkažejo grafovske ponazoritve, v katerih vozlišča, označena s prostorsko-časovnimi značilkami, predstavljajo igralce na igrišču, njihove povezave pa interakcije med njimi. Grafi se za uporabne izkažejo tudi pri modeliranju spremenljivega števila igralcev, kar olajša ravnanje s podatki, vzorčenimi iz video-prenosov z omejenim vidnim poljem.

Ob odločitvi za ponazarjanje trenutkov z nogometnih tekem s pomočjo grafov se za pridobivanje želenih vpogledov ponudi metoda strojnega učenja z grafovskimi nevronskimi mrežami (GNN) – razredom induktivno močnih, sodobnih globokih modelov, z dobrimi rezultati iz različnih domen [10–13]. Prav tako so GNN pri svojem učenju ekvivariantne na rotacije, permutacije, translacije in preslikave na vhodni ponazoritvi. Ta lastnost se za reševanje

problemov, povezanih s športom dveh ekip, ki igrata ena proti drugi na simetričnem igrišču, izkaže za zelo koristno.

V tem magistrskem delu raziščemo zmožnost modeliranja podajalčevega izbora tarče med nogometno tekmo. Naš cilj je primerjati več različnih načinov za izgranjo grafovskih ponazoritev uspešnih podaj na nogometni tekmi in oceniti vpliv različnih gradnikov procesne verige (od arhitekture nevronske mreže, do izbire podmnožic uporabljenih grafovskih značilk) na končno učno uspešnost pripravljenih modelov. V delu se posebej posvetimo predstavitvi celovite metodologije za izdelavo grafov iz odprtih podatkov. Izdelane grafe objavljamo v obliki odprtokodne podatkovne zbirke, namenjene nadaljnjim raziskavam na področju analize nogometnih omrežij.

I Kratek pregled sorodnih del

Vpliv omrežnih lastnostih na uspeh ekip v kolektivnih športih je dobro raziskano področje v domeni analize družbenih omrežij (SNA). Enega izmed temeljnih del na tem področju avtorji zaključijo s tezo, da so majhna realna omrežja najboljše pojasnjena z uteženimi povezavami med vozlišči igralcev [14]. Nadaljnje študije [15] so potrdile pozitivne korelacije med uspehom ekip ter omrežno povezanostjo, gostoto in koeficienti gručanja njihovih grafovskih ponazoritev – spet druge raziskave pa so uspešno preučevale igralne sloge ekip, fizično obremenitev posameznih igralcev in vključenost posameznikov v igralne vzorce z uporabo grafov [16–18].

Po drugi strani so globoke nevronske mreže v nogometni analitiki uporabljene že vrsto let – z njimi strokovnjaki oblikujejo ustaljene mere, kot so kakovost strela [19] in merijo potenciale napadalnih [20] ter obrambnih akcij [21]. Ti pristopi pogosto še vedno ne temeljijo na grafovskih predstavitvah latentnih ekipnih taktik. Grafi se za napovedno uporabo pojavljajo zgolj v določenih delih, predstavljenih na konferencah domenskih strokovnjakov [22, 23]. Avtorji večine tovrstnih del v njih ne uporabljajo najsodobnejših grafovskih modelov [10, 12, 13]. Raba uveljavljenih arhitektur, izkoriščujočih

ekspresivno moč mehanizma večglave pozornosti [24], ostaja precej redka. Izjema temu pravilu je nedavno delo [25], ki prav tako ne uporablja arhitekture transformerja [26, 27], uporabljenega v našem delu. Še ena izmed razlik med omenjenim delom in našo metodo je bogatenje izdelanih grafov z značilkami vozlišč in povezav. Uporabnost značilk analiziramo v temeljiti ablacijski študiji, v kateri tudi primerjamo našo, alternativno arhitekturo GNN, s tisto, predlagano v omenjenem delu. Nad predlagano arhitekturo pokažemo izboljšavo v vseh pregledanih učnih nalogah.

Nobeno izmed pregledanih del svojih rezultatov ne poroča na odprtih podatkovnih zbirkah. Posledično tudi rešitve izzivov, s katerimi se ob delu z njimi lahko srečamo, niso korenito dokumentirane. Vse omenjeno predstavljamo kot dodaten doprinos dotičnega raziskovalnega dela širši skupnosti raziskovalcev na področju SNA.

II Predlagana metoda

Podatkovno zbirko grafov, ki predstavljajo situacije na nogometnih tekmah v času oddaje uspešnih podaj sestavimo z združevanjem dveh podatkovnih nizov – dogodkovnega niza z informacijami o podaji, in niza, pridobljenega iz slikovnih okvirjev lokacij igralcev. Končna podatkovna zbirka je sestavljena iz grafov $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ z značilkami vozlišč $\mathbf{X}$, sosednostno matriko $\mathbf{A}$ in značilkami povezav $\mathbf{E}$. Za vsako vključeno podajo predstavimo tri alternativne grafske konfiguracije, ki se razlikujejo v načinu določitve sosednosti – $\mathbf{A}$ in številu vključenih vozlišč $|\mathcal{V}|$. Prva sledi obliki povezljivosti vozlišča z naperami (angl. *hub and spokes*), v kateri so povezani le podajalec in njegovi soigralci. Drugi obliki sta polno-povezana grafa – prva vključuje le podajalčeve soigralce, druga pa tudi nasprotnike. Vse konfiguracije uporabljajo iste značilke vozlišč in povezav, med katere štejemo indikatorje ekipe pripadnosti in igralčeve koordinate.

Da med označbe vozlišč vključimo tudi smerne vektorje gibanja igralcev ob trenutku podaje, vsak zamrznjeni posnetek povežemo z najkasnejšim pred-

hodnim posnetkom dogajanja na tekmi. Igralce, ki so v podatkovni zbirki anonimni, med posnetkoma nato ujemamo z uporabo madžarskega algoritma, minimizirajoč skupno prepotovano razdaljo vseh igralcev na igrišču [28]. Pri igralcih brez jasnih ujemanj uporabimo ničelne trajektorije premikanja, vratarje pa obravnavamo ločeno.

V grafovske predstavitve vključimo tudi značilke, ki bolj eksplicitno modelirajo latentno stanje posesti na igrišču. Za dosego tega uporabimo model nadzora nad igriščem iz pregledane literature [29]. Gre za evalvacijo verjetnosti, da bo dana ekipa uspela doseči žogo na določenem mestu igrišča pred tekmo. V delu predstavimo dve implementaciji algoritma za določanje nadzora – preprosto Voronoijevo teselacijo in Spearmanov model nadzora, ki upošteva reakcijski čas in hitrosti igralcev.

III Modeliranje in eksperimentalna evalvacija

Naša končna podatkovna zbirka vsebuje 80,332 grafovskih reprezentacij iz 166 tekem, osredotočenih na uspešne podaje. Analiza kaže, da so v zbirki nadpovprečno zastopane podaje obrambnih igralcev, in da so te pogosto lateralne narave. Njihovo uporabnost evalviramo na treh eksperimentalnih nalogah.

Prva naloga je regresija koordinat uspešne podaje, drugi dve nalogi pa sta klasifikacijski; napovedovanje vloge prejemnika podaje in napovedovanje ciljne cone podaje glede na območja nogometnega igrišča. Iregularno delitev igralnega polja prevzamemo po eni izmed implementaciji pozicijske igre; specifičnega taktičnega modela, ki ga uporabljajo trenerji kot Johan Cruyff, Hansi Flick, Tomas Tuchel in Pep Guardiola [30].

Vsi modeli uporabljajo enak nabor značilk vozlišč in povezav. Osnovani so na hrbtenici GNN, s katero modele najprej naučimo izdelave latentne matrike značilk vozlišč $\mathbf{H} = f_{\text{GNN}}(\mathbf{X}, \mathbf{A}, \mathbf{E})$, ki jih z namenskimi napovednimi glavami nato združimo in dodatno transformiramo za reševanje posameznih nalog. Modele treniramo z uporabo stohastičnega gradientnega

spusta (SGD) – natančneje z algoritmom Adam [31], pri čemer za regresijo minimiziramo povprečni kvadratni odklon glede na Evklidske razdalje, za klasifikacijo pa minimiziramo navzkrižno entropijo. Slednjo v primeru napovedovanja vloge prejemnika podaje pred rabo mehke maksimizacije napovednega vektorja (*angl. SoftMax*) še maskiramo glede na vloge, prisotne na igrišču v trenutku podaje. Tako model nikoli ne bo napovedal vloge, ki glede na veljavne taktične napotke, ni zastopana na igrišču.

Arhitektura naše GNN temelji na tehniki posredovanja sporočil med vozlišči in njihovimi sosedi, pri čemer uporabljamo večglavi grafovski transformer [27]. Zaradi negotovosti pri označevanju prejemnika podaje smo napovedne naloge postavili na raven grafa in ne vozlišč – zato pred razčlenbo arhitekture modela na poljubno število napovednih glav matriko $\mathbf{H}$ povprečimo po vrsticah. Med treniranjem uporabljamo postopno zmanjševanje učne hitrosti in modele treniramo na 80/20 odstotni delitvi podatkov med učnim in testnim nizom, negotovost rezultatov pa poročamo z uporabo 3-kratnega navzkrižnega preverjanja.

IV Rezultati in ablacijska študija

Tabela 1 prikazuje rezultate predlaganega modela in njegovih izpeljank na regresijski nalogi napovedi koordinat uspešnega sprejema podaje. Z ablacijsko študijo pokažemo, da dodajanje značilk s trajektorijami premika in dodajanje vozlišč igralcev nasprotne ekipe doprineseta k napovedni točnosti. Prav tako je opazna rahla izboljšava napovedne točnosti ob dodajanju značilk, povezanih z nadzorom igrišča. Predlagani model napram ostalim pregledanim hrbteničnim konfiguracijam GNN, dosega najboljše rezultate. Podobne izsledke opazimo tudi pri učenju na klasifikacijskih problemih, kjer najboljši model pravilno napove območje pozicijske igre v 55 ± 2 % (Točnost@3 = 85 ± 1 %), pravilnega soigralca pa v 42 ± 1 % (Točnost@3 = 73 ± 1 %) primerov. Empirično model ocenjujemo za uporabnega. V delu pokažemo, da napovedna natančnost pada pri podajah s kril, narašča pa v primeru podaj

Model	Sr. vrednost evklidske napake [jardi]
Predlagani model (p. m.)	$6.78 \pm .21$
GATv2 hrbtenica [24, 25]	$7.80 \pm .10$
p. m. brez vektorjev premika	$9.14 \pm .29$
p. m. brez nadzora igrišča	$7.08 \pm .06$
p. m. brez nasprotnikov	$7.89 \pm .08$
p. m. – konfiguracija vozlišče z naperami	$7.92 \pm .08$
GCN hrbtenica [12]	$15.67 \pm .05$
Globoka agregacija nizov [32]	$12.04 \pm .04$

Tabela 1: Povzetek tabele 5.2, ki prikazuje rezultate ablacijske študije na različnih modelih in primerjavah. Interval zaupanja prikazuje standardni odklon distribucije napake treh modelov v sklopu prečnega preverjanja.

v napadalno tretjino, in v primeru dolgih podaj oz. oddaljenih ciljev.

Chapter 1

Introduction

In modern association football (henceforth, football), data-driven insights play a key role in refining match preparation and formulating counter strategies on a match-by-match basis, with clubs increasingly employing data professionals to support coaching and the development of game plans [2]. Aggregated insights and visualizations of advanced metrics have also become a key factor in viewers' choice of sports broadcaster, and are reported to enhance the viewing experience for those with less familiarity with the sport [3, 33]. The use of these metrics is highly prevalent in betting markets as well [4]. In short: data and insights, generated either through aggregation, or training specialized models, are being utilized in virtually every aspect of the game.

The aforementioned insights are generated by using data, made available in different formats. Aggregating traditional streams of tabular data, which often capture spatiotemporal and descriptive aspects of individual players, remains a common approach. These streams are typically generated from tracking data gathered by specialized sensors, or by sampling the (broadcast or proprietary) video feeds [5–7]. Nevertheless, the fluid and dynamic nature of the game poses significant challenges in breaking down the action into discrete events for eventual analysis [8]. One promising approach to modeling the spatiotemporal data of any given moment on the pitch is through the use of graphs. Graphs do not only represent individual players (as nodes)

but also capture the relationships between them — relationships that have been shown to be crucial for certain analytical tasks [9]. This representation format is also robust against some common negative patterns, encountered in the raw data. One such pattern is the variable number of players visible in a match snapshot due to limitations in broadcast camera angles. While one would have to resort to suboptimal methods of representation in such instances (i.e. padding a fixed-length feature vector) when using other approaches, graphs can simply be adapted in their number of nodes. Furthermore, established social network analysis (SNA) metrics have been shown to correlate with play-patterns and frequencies of certain on-pitch events, providing valuable perspectives by simply analyzing the graph representations of match moments [15–18].

Another reason for the promise of this method of event modeling is its compatibility with graph-based machine learning models, such as graph neural networks (GNN). Not only does this class of model boast a plethora of best-in-class results on various tasks [10–13], it is also equivariant to rotations, translations, reflections and permutations in the input data [34] – a property that is very desirable in the domain of modeling two-team head to head sports with a symmetrical playing field.

In this thesis, we limit our exploration to a specific subdomain of football – modeling the decision-making of a football player when making a pass to progress the ball on the field while retaining possession. Our aim is to explore whether modeling the pass-moments using graphs results in representations, useful for downstream tasks. If so, we look to explore how can one create these graphs to achieve the most promising predictive results. Special care is taken to introduce a comprehensive methodology to construct such graphs from open data, so that they can be made available as a dataset for further research as a deliverable of this thesis.

In short, this thesis sets out to:

- overview the openly accessible football match data;
- introduce and overcome challenges that arise with using such data to create graphs for downstream machine learning (ML);
- select the GNN architecture, most appropriate for the selected ML objectives;
- evaluate the different manners of graph construction using the selected architecture;
- share insights about the model behavior and offer a reproducible and reusable graph dataset for future work by the SNA community.

1.1 Related Work

Team performance in collective sports has become a well-researched area in the SNA community. The work in [14] is a good starting point for the study of this field. It suggests that small real-world networks are best explained using weighted edges between nodes, representing individual players. The work also introduces custom SNA measures that are analogous to those found in more general work on larger, unweighted networks, and directly connects them to team performance. Subsequent authors [15] also studied and confirmed positive correlations of network connectivity, density, and clustering coefficients with team success. Others were able to extrapolate even more interesting findings, such as play styles of teams [16], the amount of physical strain on specific players [17], and the play-making involvement of individuals [18], directly from these sorts of graph representations. While representing team dynamics with player nodes is most prevalent in the literature, authors have also had success in modeling football in different ways; i.e. as an interaction network of different pitch regions [35].

Many deep learning (DL) scholars, working on problems relating to football, have concerned themselves with establishing metrics that describe the qualities of different on-field events. These include shot quality (such as the expected goals metric – often abbreviated as xG) [19] and the potential of both attacking [20], as well as defensive actions [21]. Most of these approaches take geospatial data of individual actions into account but fail to incorporate the rich graph representations of latent team tactics that might be in play at the sampling time of any observed event datum.

Some approaches, published by footballing science practitioners, have used rich tracking data to construct graph representations for downstream predictive use. In [22], the authors create general-purpose graph embeddings of football passes. They train these by classifying graphs into classes, denoting all possible combinations of pass destination buckets, body parts used, and pass types. They also show the applicability of these general-purpose embeddings for domain experts’ ability to identify phases of play. Authors of [23] focus on predicting the likelihood of individual players receiving passes or shooting during attacking possessions. While both of the described approaches use geo-spatial graphs at input, the model architectures, described therein, fail to include recent advances in graph deep learning. Namely, the used architectures either omit message passing of modern graph neural network (GNN) modules entirely [10, 12], favoring splitting the node and edge features into several multi-layer perceptron (MLP) blocks, or use modules that are proven to not be able to distinguish certain simple graph structures [13].

In the presented work, we instead experiment with architectures that leverage multi-head attention [24], as well as the established well-performing architecture of the transformer [26, 36] adapted to the domain of graphs, as presented in [27]. Such methods have been shown to perform well in the domain of multi-agent tracking prediction in sports. Such methods, often using fully connected networks of graph nodes, have been proven to outperform standard convolutional networks in the fields of player trajectory

prediction [6, 37] and evaluating passable regions from the teams' positioning [38].

Only scarce attempts have been made to analyze on-ball events in football matches by leveraging modern graph neural networks. One recent work [25] attempts to understand corner kicks (whilst stating the ability to generalize to other set pieces), by predicting pass receivers, evaluating the likelihood of a shot being taken after the set piece, and generating the evolution of the player formations by regressing the nodes' geo-spatial coordinates.

While similar, the methodology of graph construction in these works fails to include any rich node and edge features, opting to keep the nodes anonymous and to not describe momentary formation attributes in the players' vicinity explicitly to the model. We analyze the impact of including such features in our graph construction. We also opt for alternative graph-attention operators when compared to those presented in the cited work, and additionally extend the authors' efforts to the domain of regular play passes.

Finally, none of the reviewed work uses openly accessible datasets, nor do the authors present solutions to challenges that arise when working with such sources. No ready-made graph representations of football match events and player positioning are openly available. We argue that this is a solid motivation to not only study this field but also prepare a resource for future work in the domain.

1.2 Structure

This thesis proposes a geometric deep learning pipeline for the preparation, analysis and evaluation of pass-moment representations with graphs. A main deliverable of our work is a dataset of 80,332 graphs, and a collection of graph neural network models, trained for several downstream tasks, built upon said representations. The steps taken to construct and analyze these are presented in sequence as follows: Chapter 2 overviews the structure and limitations of the open-source football event datasets from which the graphs

are constructed. The construction methodology, along with the steps taken to circumvent those limitations, is introduced in Chapter 3. Its contents then continue into a thorough analysis of the graph datasets that we deliver as part of this thesis. The utility of the graphs and their features is benchmarked on several predictive tasks: regressing the pass destination, classifying the pitch zone according to the map of positional play, and classifying the pass recipient. We introduce the methodology of our modeling and formalize the selected architecture in Chapter 4. We then present the results of our best-performing models in Chapter 5 and analyze the contribution of different graph structures and features in an ablation study in Section 5.1. We conclude with a summary of our findings and outline the paths for future work in Chapter 6.

Chapter 2

Football Match and Player Data

To construct graphs of football match events, we utilize the open football match event dataset, provided by the sports data company Statsbomb (henceforth, the provider) [39]. Along with the raw data files in the cited repository, the provider makes the event stream interfactable through the *Statsbomby* package for the Python programming language, available through the pip package installer [40].

The provider offers two different sets of information for our desired granularity. The first is a detailed log of on-ball match events with spatio-temporal information regarding the actor (i.e. the passer, the receiver, the player making a carry, shot, or save) and the event in question. We refer to this collection as *event data*. The second stream is joinable to the first in a many-to-one fashion (via the *id* columns in Tables 2.1 and 2.5), and denotes the coordinates of all on-screen players at the time an event happens. The spatio-temporal information is given in the coordinate system, introduced in Figure 2.1. These data points are referred to as *360 frames* by the provider’s branded nomenclature. In this thesis, we refer to them as *freeze frames*, to retain generality.

As both sets are required to construct the graphs in our dataset, we

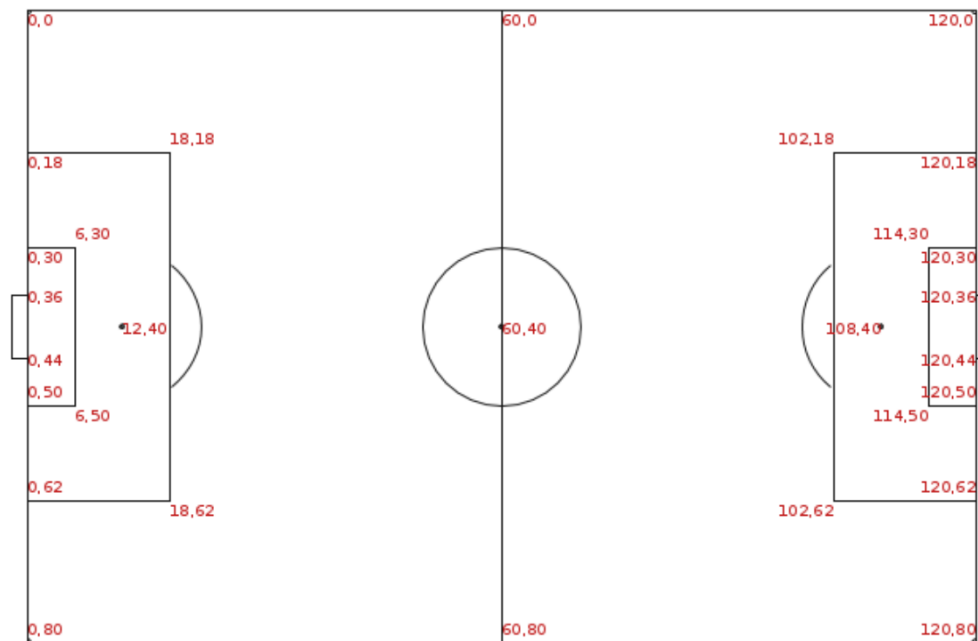


Figure 2.1: 2D Coordinates of key locations, as visualized by Statsbomb.

limit ourselves to the collection of competitions, offering both sets for male international football matches openly¹ – the 2022 FIFA World Cup, and the 2020² and 2024 iterations of the UEFA European Championship.

Given our interest in modeling passes, each pass from any of the 166 matches in the three competitions becomes a candidate for graph construction. In the remainder of this chapter, we introduce the structure of each tabular dataset and the utilities of their respective columns in more detail.

2.1 Events

A general overview of the information, available for all events, and relevant for our downstream graph construction, is available in Table 2.1. Additional information, available only for pass events, is tabulated in Table 2.4.

¹A larger collection of free event data is offered without matching freeze frames

²These matches were actually played in 2021 due to the COVID-19 pandemic.

We augment each pass event’s row with the formation of players, present on the fields at its timestamp. We do this by joining the pass event’s timestamp to the most recent *Tactical Shift* event, adhering to replacements made by *Substitution* events succeeding it. These meta event types are briefly introduced in Table 2.2. We also filter out any passes for which *Body part* from Table 2.4 is not one of the players’ feet, or the *Outcome* was of any known value (i.e. incomplete, out, offside, or an injury clearance). Finally, we annotate each pass with its most recent preceding known event that can be joined to a freeze frame. The frames from both events will be utilized for calculating player trajectories, discussed in Section 3.2. Finally, we filter out any passes with no available preceding event in a 3-second window, for the sake of preserving realistic player trajectory calculations.

2.2 Freeze Frames

A row in the freeze frame dataset comes as a triplet of an id, linking to its associated event in the event stream, a *visible_area* array of coordinates, and the freeze frame itself. The *visible_area* describes a polygon on the pitch that was visible to the camera at the time the freeze frame was collected. The polygon is structured as a sequence of coordinates $L = [X_1, Y_1, X_2, Y_2, \dots, X_n, Y_n]$, in the pitch coordinate system, as shown in Figure 2.1. Such polygons are visualized later in the work – for example in Figures 3.2, 3.3 and 3.5. The set describes a closed loop on the pitch, with (0,40) denoting the center of the actor team’s goal in any frame, meaning that subsequent events associated with opposing teams will be associated with a mirrored loop:

$$L' = [X_{n'} = 120 - X_n; Y_{n'} = 80 - Y_n; \forall (X_n, Y_n) \in L]. \quad (2.1)$$

The freeze frame includes a row for each player in the visible area. The available information for each player is tabulated in Table 2.5. Notice that there is no identifier of individual players available, only team affiliation and whether or not the player is the goalkeeper. Also note that, when comparing

Column	Description	Value example
Id	The unique identifier for each event.	0052d1b5-e2b0-4629-bbea-c18c884ab103
Period	Part of the match that the event takes place in.	1=1st half (regulation) 2=2nd half (regulation) 3=1st half (extra time) 4=2nd half (extra time)
Timestamp	Time within period, recorded in milliseconds.	00:01:06.293
Minute	The minutes on the clock. Resets to 45 at half-time, 90 at the start of extra time, etc.	46
Second	The amount of seconds in the timestamp.	6
Type	Event type name.	See Table 2.2.
Possession	The unique period of play in which the ball is in play and a single team is in control of it.	Ordered sequence of natural numbers
Possession team	Id of the team that started the current possession and is in control of the ball – even if the event in question relates to the opposition.	243 / "Senegal"
Pattern	Name of the event-associated play pattern.	See Table 2.3
Team	Name of the team this event relates to (not necessarily the team in possession).	512 / "Belgium"
Player	Name of the player this event relates to (only if event is tied to a specific player).	5079 / "Zlatan Ibrahimovic"
Position	Name of the position the acting player was in at the time of this event.	1 / "Goal Keeper"
Location	Array containing two integer values; the (x,y) coordinates of the on-ball event.	(60,40) for the center of the pitch.
Duration	Duration of event, in seconds, if applicable.	2.152
Related events	The ids of related events – a shot might be related to the goalkeeper event and a block event. A successful pass is related to a pass receipt.	[1e52d1b5-e2b0-4bb9-b66ea-c18c2584ab1423,...]
Tactics	This object is added for <i>Starting XI</i> and <i>Tactical Shift</i> meta-event types, as seen in Table 2.2. It describes the formation being used and lists all the players' roles within it.	{433: {"Goalkeeper": "Hugo Lloris", "Right Back": "Jules Koundé", ... }}

Table 2.1: General overview of data available for all events

Event Type	Description
Ball Receipt	The receipt or intended receipt of a pass.
Ball Recovery	An attempt to recover a loose ball.
Dispossessed	Player loses ball to an opponent without attempting a dribble.
Duel	A 50/50 contest between two players of opposing sides.
Camera On*	A stop of the camera capturing the gameplay, to showcase a replay.
Block	Blocking the ball by standing in its path.
Offside*	Cases of offside infringement resulting from a non-pass event.
Clearance	Action by a defending player to clear the danger without an intention to deliver the ball to a teammate.
Interception	Intercepting pass from recipient by moving into passing lane.
Dribble	An attempt by a player to beat an opponent.
Shot	An attempt to score a goal with any legal part of the body.
Pressure	Applying pressure to receiving, carrying or passing opponent.
Half Start	Signals referee whistle to start a match period.
Substitution	Denotes a player leaving the pitch in favor of a replacement.
Own Goal Against*	An own goal scored against the team.
Foul Won*	A player wins a free-kick or penalty for their team.
Foul Committed	Any non-offside infringement that is penalised as foul play.
Goal Keeper	Any action that can be done only by the goalkeeper.
Bad Behavior*	A player is carded due to a non-play-related infringement.
Own Goal For*	An own goal scored for the team.
Player On*	A player returns to the pitch after a Player Off event.
Player Off*	A player goes/is carried out of the pitch without a substitution.
Shield	Player shields ball from opponent, leaving it to go out of bounds.
Pass	Ball is passed between teammates.
50/50	2 players challenging to recover a loose ball.
Half End*	Signals the referee whistle to finish a match part.
Starting XI*	Meta event, indicating the players in the starting 11 with their position and the teams' tactical formation.
Tactical Shift*	Formation change – shows the players' new positions.
Error	Player is judged to make an on-ball mistake that leads to a shot.
Miscontrol	Player loses ball due to a bad touch.
Dribbled Past	Player is dribbled past by an opponent.
Injury Stoppage*	A stop in play due to an injury.
Referee Ball-Drop*	Referee drops the ball to continue the game after an injury stoppage.
Carry	A player controls the ball at their feet while moving.

Table 2.2: Showcase of different event types. While our dataset only corresponds to formations at the time of *pass* events, all events, except for those labeled with (*) are sampled as possible *preceding events*.

Play Pattern	Description
Regular Play	Not part of any of the special play patterns below.
From Corner	Passage of play following a corner.
From Free Kick	Passage of play following a free-kick.
From Throw In	Passage of play following a throw-in.
From Counter	Passage of play, where the possession started with an open play turnover outside the counter-attacking team's final third and the attack traveled at least 18 yards towards the goal.
From Goal Kick	Passage of play following a goal kick.
From Keeper	Passage of play following any non goal-kick keeper distribution.
From Kick Off	Passage of play following a break or goal.

Table 2.3: Play patterns, included in the source dataset

Attribute Name	Description	Value Example
Recipient	Name of the recipient of the pass, or for whom an incomplete pass was intended.	2934/"Erik Lamela"
Length	Distance of pass in yards.	16.03
Angle	Angle of pass in radians – 0 pointing straight ahead, positive values in clockwise direction.	-2.49
End Location	Coordinates of pass end.	[100, 45]
Body Part	Name of body part used to make this pass.	Drop kick, Head, Keeper arm, Left foot, Right foot, No touch, Other
Type	Attribute option describing the pass type.	Corner, Free Kick, Goal Kick, Interception, Kick Off, Recovery, Throw-In
Outcome	Category of pass' special outcome (if any).	Incomplete, Injury Clearance, Out, Pass Offside, Unknown

Table 2.4: A table of additional information for pass events.

Field Name	Description
Event ID	Unique identifier for the matching event from Table 2.1.
Visible Area	Array of coordinates describing the closed loop polygon, visible to the camera, from which the freeze frame is collected; formatted as $[X_1, Y_1, X_2, Y_2, \dots]$. Player locations may be outside of the visible area where these were manually placed.
Location	X, Y position of a player on the field, with coordinates oriented in the same direction as the linked event.
Teammate	Indicates whether the player plays on the same team as the event's <i>actor</i> .
Actor	Indicates the current player as the one performing the associated event.
Keeper	Indicates this player is a goalkeeper.

Table 2.5: A description of the freeze frame set of data, joined to Table 2.1 as part of our preprocessing.

the frames of two events, as discussed at the end of Section 2.1, they can be associated with on-ball events of different teams. In that case, the players' team affiliations in the freeze frame, as well as their coordinates, need to be reversed/mirrored, to retain consistency between the data points upon comparison.

We apply such preprocessing when joining each pass event's freeze frame, as well as the freeze frame of its most recent preceding event, to the pass event's row in the event stream. It should also be noted that not all 22 players are necessarily visible in any given frame's visible area. For the sake of the generality of information in our dataset, we filter out any frames where the passing player's team does not have at least 3 other players in the camera's view.

The bounds of the freeze frame dataset are given as a fixed 80 yards of pitch width and 120 yards of pitch length. Consequentially, all distances in this thesis are reported in yards, to retain consistency between the dataset and the results, reported in this document. When discussing distances in the thesis body, conversions to meters are provided. A fixed conversion of 0.9144 meters per yard is used.

2.3 Player Data

Supplementary work to this thesis includes efforts of player identification. Not only do we hypothesize that accurately labeling players in otherwise anonymous freeze frames could aid in downstream machine learning tasks (and conduct experiments to evaluate whether our method is appropriate to drive these improvements); but accurately labeling the players’ names and positions in graphs would also be instrumental for their usage for analytical & visualization purposes. We use the dataset of player attributes from the FIFA³ series of games, made available on the sofifa.com repository [41–43]. More information about this dataset, our methodology for matching it to the Freeze Frame data, and its effect on predictive performance can be found in Appendix A. An overview of the used features is shown in Table 2.6.

Attribute	Value Range
Short Name Long Name Nationality Name	String values, used for matching to Table 2.1
Value EUR	0 – 190.5 cm
Wage EUR	0 – 560k
Age	16 – 53
Height CM	155 – 206
Weight KG	49 – 110
Preferred Foot	Left / Right
Detailed Attributes	0 – 99, for more see Table A.1.

Table 2.6: Attributes, relevant for generating node features from the FIFA / EA FC video games. The first section corresponds to the features, required to match player names between the datasets. Other attributes correspond to those used either explicitly or implicitly (dimensionally reduced as described in Appendix A) as node features in our player-annotated dataset.

³Rebranded as EA FC in the 2024 iteration, used for labeling the passes from UEFA EURO 2024.

Chapter 3

Graphs & Construction Methodology

In general terms, our dataset consists of graphs $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, parametrized with sets of nodes $\mathcal{V}$ and edges $\mathcal{E} \subseteq \mathcal{V} \times \mathcal{V}$. We denote the adjacency of two vertices (and thus the existence of an edge between them) using the adjacency matrix $\mathbf{A}$. The graphs are featurized with $\mathbf{X} \in \mathbb{R}^{|\mathcal{V}| \times f}$ denoting the matrix of node features and $\mathbf{E} \in \mathbb{R}^{|\mathcal{V}| \times |\mathcal{V}| \times g}$ denoting the matrix of edge features, where f and g are the cardinalities of the sets of independent features, attributed to each node and edge in any given graph.

In the following sections, we initially introduce the different methods of determining $\mathbf{A}$ (i.e. the graphs' neighborhood structure), and then expand upon our methodology for featurizing them using match-specific information, ascertained from the on-field situation at the time of the freeze frame's capturing. In section 3.4, the dataset is analyzed, and aggregated insights about its general properties are visualized. The three graph variations that we generate for each pass are shown in Figure 3.1.

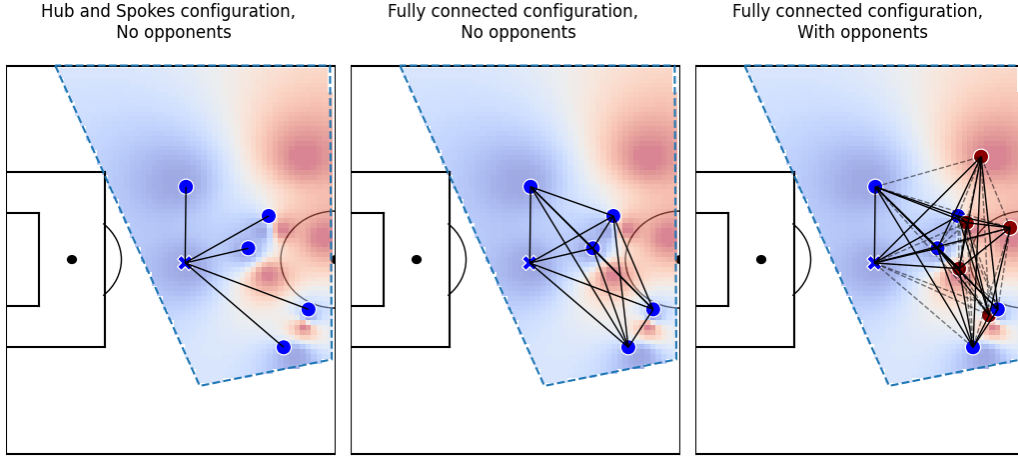


Figure 3.1: Comparison of the three graph adjacency configurations. In the right subfigure, the edges with the false *'is_teammate'* feature are drawn with dashed lines, whereas the nodes are colored according to team affiliation. The actor is indicated with a cross. The background shows an example of the Spearman pitch control field, explained in detail later in this chapter.

3.1 Graph Structure

In the dataset, we present three graph configurations, all available with the same node and edge features, tabulated in Table 3.1. The first variation uses a *"hub and spokes"* configuration, with only the actor (passer) and his teammates present in the graphs (as nodes), with all teammates connected to the actor and no other node, i.e. $\mathcal{E} = \{(a, v) \mid v \in \mathcal{V} \setminus \{a\}\}$. There, a denotes the actor's node representation. The two other graph configuration methods correspond to fully connected graphs ($\mathcal{E} = \mathcal{V} \times \mathcal{V}$), with one's nodes corresponding to only the passer and his teammates, and the other also modeling the opposing players, present in a given freeze frame. For all configurations, we make the same node and edge features available, with the graph denoting players from both teams also possessing an edge feature, indicating whether the connected nodes belong to the same team.

Feature Name	Range	Description
Node features		
Is Teamate		
Is Actor	$\{0, 1\}$	See Table 2.5.
Is Keeper		
Location X	$[0, 120]$	Player's current location along the axis.
Location Y	$[0, 80]$	
Trajectory X	$[0, 13)$	Normalized trajectory & velocity the axis, calculated from preceding frame.
Trajectory Y		
Personal space control*	$[0, 1]$	Fraction of the theoretically possible pitch control in the 3/8-yard radius centered around the player.
Edge features		
Edge Control Homogeniety*	$\{0, 1\}$	Whether the edge travels over a path that changes the majority PC team more than once.
Fraction of Edge Control*	$[0, 1]$	Fraction of the theoretically possible PC across the direct link between the players.
Distance	$[0, 144]$	Explicitly states the distance between the players.
Teammate indicator	$\{0, 1\}$	Indicates whether the connected players play on the same team.

Table 3.1: Overview of node and edge features, made available in our dataset. Features annotated with (*) have two variations, according to two different methods of calculation, expanded upon in Section 3.3.1.

3.2 Player Trajectory Vectors

Much of the related work utilizes player trajectory vectors in addition to the players' absolute positions on the pitch for downstream machine learning tasks [5, 7, 20, 25]. Indeed, when evaluating a player's ability to receive the ball in a certain area of the pitch, the trajectory of their movement at the moment the pass is taken will influence their ability to arrive at its destination at the correct time. However, as the players in the Freeze frame dataset from Section 2.2 are anonymous in any given pitch snapshot, accurately calculating their velocity and trajectory becomes non-trivial.

As discussed in Section 2.2, we associate each frame with the most recent preceding on-ball event that has a frame associated with it and ensure that the team features are uniform across both frames. This corresponds to mirroring some of the frames in instances where the associated events have opposing actors – as the frame is always expressed in the orientation of the actor's attack. We note that the previous rule does not hold in the entirety of the source dataset, so we apply additional quality control by calculating the per-team centroids of the paired frames and perform mirroring only in cases where it reduces the cross-frame team centroid distance.

After suitable preprocessing, we pose the identification of coordinate pairs as an optimization problem as follows:

Let n correspond to the amount of nodes in a given pass freeze frame and m correspond to the amount of nodes in its associated preceding frame. Then, let C be an $n \times m$ matrix representing the distances of each point in the pass freeze frame to each point in the associated preceding frame;

$$C = \begin{bmatrix} c_{11} & c_{12} & \dots & c_{1m} \\ c_{21} & c_{22} & \dots & c_{2m} \\ \vdots & \vdots & & \vdots \\ c_{n1} & c_{n2} & \dots & c_{nm} \end{bmatrix}, \quad c_{ij} = \sqrt{(p_{ix} - a_{jx})^2 + (p_{iy} - a_{jy})^2},$$

where p_i corresponds to the i -th player present in the pass freeze-frame and a_j to the j -th player in the associated preceding freeze frame. The x and y

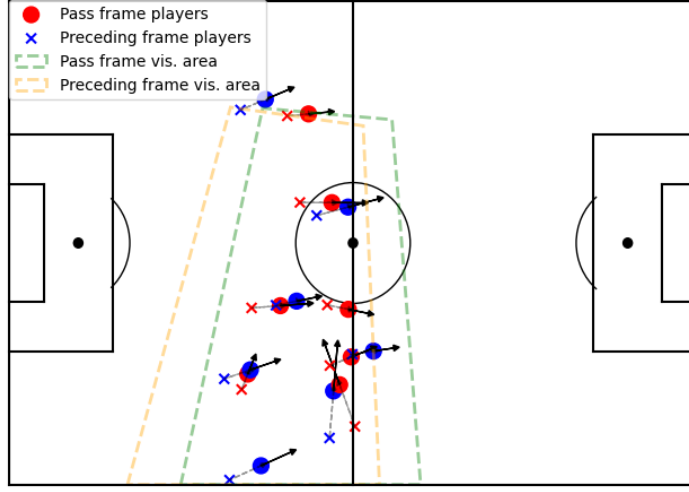


Figure 3.2: Example mapping between player locations from two frames, with dashed lines connecting the preceding frame players to their assigned pairs in the observed frame.

indices represent the player’s coordinates along the horizontal (x) and vertical (y) axes on the pitch.

Our objective is to assign player pairs between the associated freeze-frames in a way that minimizes the total distance covered by all players; thus making the movement the most likely in a short time-frame. Since each player can only map to a single point in the preceding frame, the assignments represent an independent set of the matrix C . This can be expressed as permuting the rows of C to minimize its trace,

$$\min_P \text{Tr}(P \cdot C),$$

where P is the permutation matrix. We solve this using the Hungarian assignment algorithm [44]. We implement it using the `munkres` package for the Python programming language [28]. An example of a successful mapping of players can be seen in Figure 3.2.

Note that, in cases where the associated preceding freeze frame has more candidate players than the frame associated with the pass, some points re-

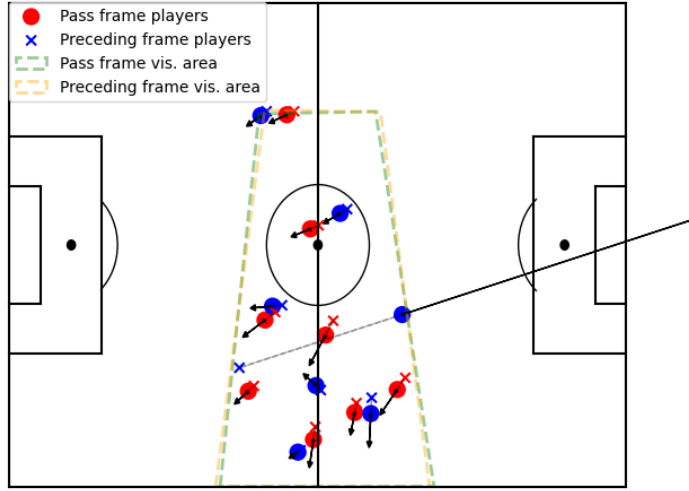


Figure 3.3: An example of an erroneous mapping due to a player on the blue team exiting the frame, while another one enters. Such extreme trajectories are removed from the final dataset (players are assumed to be static).

main unassigned. In cases where the passing frame has more candidate players present, we transpose the cost matrix and assign the player with no pair in the preceding frame a null trajectory (i.e. we assume they were static at the time of sampling).

While the trajectories for the majority of the dataset prove easily calculable in the described way, some outliers do persist, mostly due to the limitation of players leaving and entering the visible area between frames. Observe, for example, the players present in both frames shown in Figure 3.3. Our methodology would naively pair the points, obviously corresponding to players standing on opposing edges of the visible area, whereas a far more likely explanation of the positioning would be one player entering the frame, while another left it. To tackle such cases, we apply an upper heuristic threshold of 13 yards (11.89 meters) to all trajectories, nulling them in cases where a single player would be assigned one. In cases where multiple players are assigned these long trajectories, we discard the pass from our final dataset, as an additional step of quality assurance.

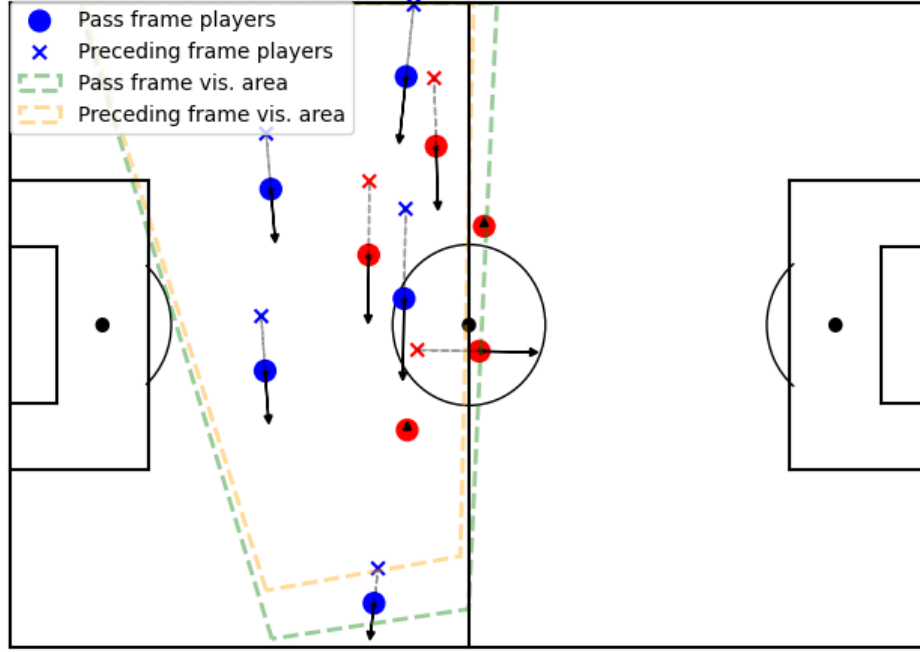


Figure 3.4: An example of an erroneous mapping due to a red player entering the frame being closer to a red player in the preceding frame, where the overall trajectory of the remaining players indicates that the unpaired player below would be the correct match. This is solved by introducing Algorithm 1.

Finally, we also first attempt to take only points present in both the preceding and current frames into account (setting the trajectories of all newly appearing players to 0), only including the others in cases where the first approach would leave some points that were present in the initial frame without a pair. This approach for selecting which players to consider when constructing the cost matrix C is summarized in Algorithm 1, and alleviates issues, such as the example presented in Figure 3.4.

For all players without special handling (i.e. setting their trajectories to zero), we assume a continued linear trajectory, which we normalize by the millisecond difference between the capturing of both frames, as shown in

Algorithm 1 Procedure for selecting players to include in the cost matrix, input to the Hungarian algorithm.

- 1: **Input:** Locations of players (current and preceding frame) $\mathcal{L}, \mathcal{L}_p$, Sets of all viewable points in the preceding frame $\mathcal{V}, \mathcal{V}_p$
 - 2: $\mathcal{L}_{\text{initial}} \leftarrow \{\mathcal{L} \cap \mathcal{V}_p\} \setminus \mathcal{L}_{\text{keeper}}$
 - 3: **if** $|\mathcal{L}_{\text{initial}}| < |\mathcal{L}_p \setminus \mathcal{L}_{\text{keeper}}|$ **and** $|\mathcal{L}_{\text{initial}}| < |\mathcal{L} \setminus \mathcal{L}_{\text{keeper}}|$ **then output:** $\mathcal{L} \setminus \mathcal{L}_{\text{keeper}}$
 - 4: **else output:** $\mathcal{L}_{\text{initial}}$
-

the discussed figures. Finally, note that goalkeepers are withheld from this calculation, as they can be clearly distinguished from other players on the field from the input data, as also noted in Algorithm 1. All the described limitations regarding a player’s presence in only one of the frames apply to goalkeepers as well.

Velocities in both directions are encoded as features of each node in the graph. They are also required for the computation of other features, namely the ones reliant on Spearman’s pitch control model, presented in Subsection 3.3.1.

3.3 Frame-Specific Features

Along with features, denoting general meta information about an individual player (i.e. their team affiliation or their preferred foot), some node and edge features strive to model the on-pitch situation more explicitly. Some of these features are fairly trivial (i.e. the distance between two players), while others require additional explanation. This latter set of features tries to offer additional insight regarding the relative safety of selecting any teammate node in the graph as a potential recipient of the pass through the notion of pitch control (PC). Pitch control relates to the idea that any point on the playing field can be scored with respect to the likelihood that a given team will be able to control the ball, should it be passed to said location.

3.3.1 Pitch control

We introduce two different pitch control models. The first is a simple Voronoi tessellation of the pitch with respect to the nearest neighbor to any given point in the visible area of the field.

Assume a space $\mathbb{S}$ of points $\mathbf{p} = (p_x, p_y) \in \mathbb{R}^2$, where $0 \leq p_x \leq 120$ and $0 \leq p_y \leq 80$, with coordinates mapped to the provider's system as introduced in Figure 2.1. Further, assume that $\mathbb{S}_f$ corresponds to $\mathbb{S}$, masked such that only points within the visible area cone of frame f are considered. Finally, assume that $\mathbb{I}_f = \mathbb{N}_0 \cap \mathbb{S}_f$ is the space of integer-coordinated visible points in a given frame.

We can define the Voronoi pitch control function $PC_V(\mathbf{p})$ as

$$PC_V(\mathbf{p}) = \max(\mathcal{L}_{\mathbf{p}_{\text{NN}}}(\mathbf{p}), 0), \quad \text{where} \quad \mathbf{p}_{\text{NN}} = \arg \min_{\mathbf{q} \in \mathcal{P}} \|\mathbf{p} - \mathbf{q}\|_2, \quad (3.1)$$

where $\mathcal{L}_p$ is a labeling function, denoting whether a point is affiliated with the passer's team (1) or not (-1), and $\mathcal{P}$ is the set of all players, positioned at coordinates $p \in \mathbb{S}_f$, and present in the freeze-frame.

The simple Voronoi spatial pitch control model has some limitations, as it assumes that the player closest to the ball at any given time will always be able to control the ball first. This model fails to incorporate player fatigue, ability, speed, and, perhaps most important of all, movement velocity at the time of the pass. These issues are addressed in the work of Spearman [29, 45], whose pitch control model has been adopted in other reviewed work as well [38]. The works propose a mapping to the temporal space, and adding uncertainty via fitting a classifier on the ground truths of which team actually ends up controlling the ball. As fitting such a model is out of scope for this thesis, we adopt the fit parameters as reported in [29] in this work. The differences in the pitch control models are visualized in Figure 3.5.

Assume a fixed reaction time t_{react} . This is the time it takes a player to react to a pass being made. Assume that, during this time, the player continues along their current trajectory at their current velocity. Upon reacting, they take the shortest path to the ball at their fixed top speed, v_{top} . We

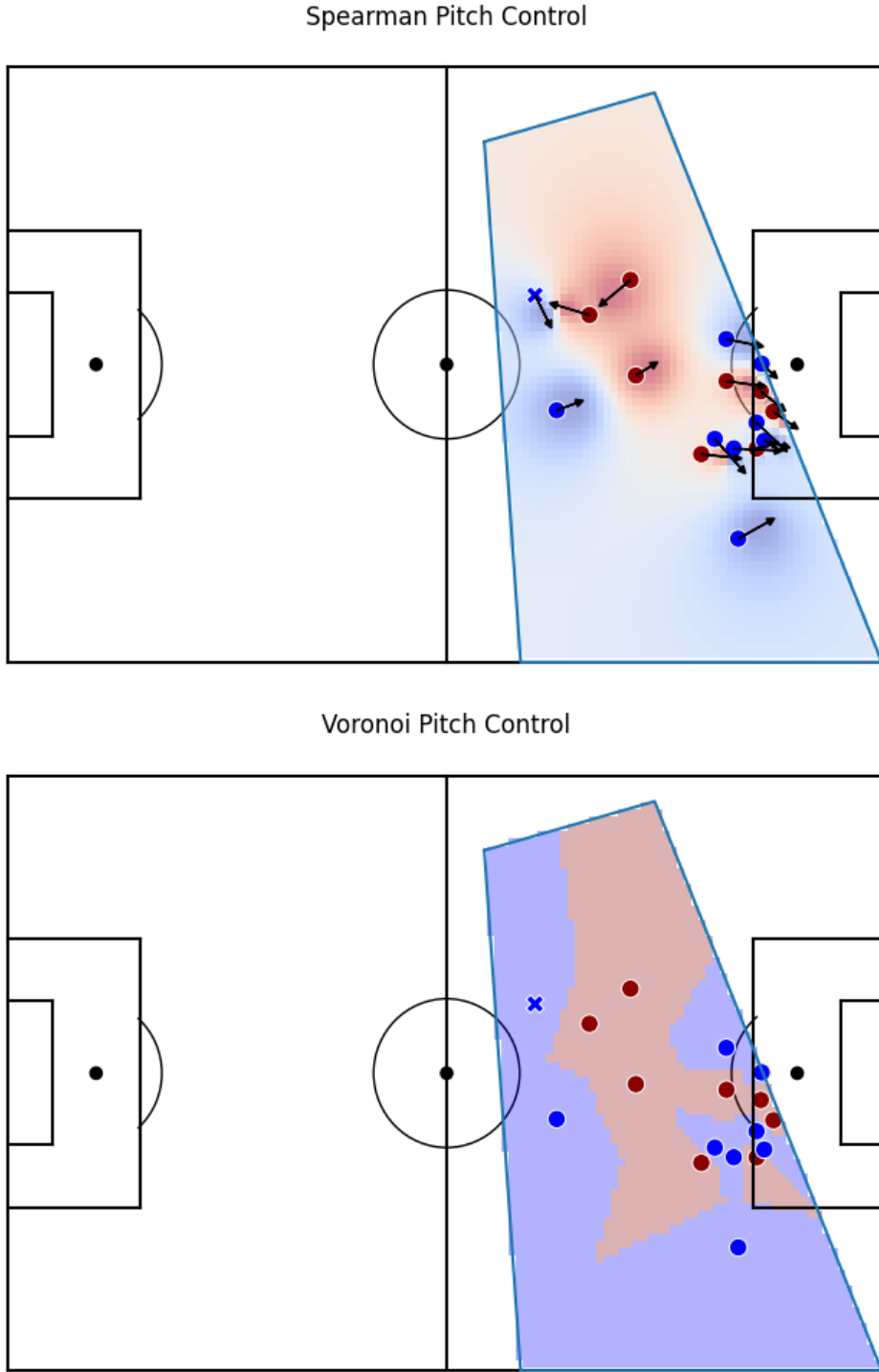


Figure 3.5: Comparison of the naive spatial pitch control model, and the temporal pitch control model, considering player velocities and ML-fit control uncertainty factor from Equation (3.2)

can define the Spearman pitch control as a function of a point on the pitch $\mathbf{p} \in \mathbb{I}_f$ as

$$PC_S(\mathbf{p}) = \left[\frac{\sum_i \mathcal{L}_i t_{i\mathbf{p}}^{-\beta}}{\sum_i t_{i\mathbf{p}}^{-\beta}} + 1 \right] / 2, \quad \text{where} \quad t_{i\mathbf{p}} = \frac{\|\vec{\mathbf{p}} - (\vec{\mathbf{q}}_i + t_{\text{react}} \vec{\mathcal{T}}_i)\|_2}{v_{\text{top}}}, \quad (3.2)$$

where $\mathbf{q}_i \in \mathbb{P}_f$ is the position of a player at the time of the pass being taken, $\vec{\mathcal{T}}_i$ is its unit trajectory vector from Section 3.2, and β is the classifier weight, to be fit in a supervised manner to minimize the sum of the squared residuals between the function value and the ground truth (1 for the passer's team, 0 for the opposition). In our modeling, we use the values of $\beta = 2.5$, $t_{\text{react}} = 0.7$ seconds and $v_{\text{top}} = 5.45$ yards (5 meters) per second. Notice that the value of PC_S now maps to the interval $[0, 1]$, and that, when using $\beta = \infty$, we get $PC_S(\mathbf{p}) = PC_V(\mathbf{p})$.

3.3.2 Pitch control features

For both pitch control models, we derive features for graph nodes and edges, as previously introduced in Table 3.1.

Fraction of edge control

We assume that the edge between two nodes corresponds to the shortest path between them. We define the edge control as the fraction of total available control on the edge, as defined by either pitch control function ($PC \in \{PC_V, PC_S\}$) as

$$EC(\mathbf{a}, \mathbf{b}) = \begin{cases} \frac{\sum_{\mathbf{q} \in \mathcal{Q}} PC(\mathbf{q})}{|\mathcal{Q}(\mathbf{a}, \mathbf{b})|}, & \text{if } |\mathcal{Q}(\mathbf{a}, \mathbf{b})| > 3 \\ 1 & \text{otherwise.} \end{cases} \quad (3.3)$$

where $\mathcal{Q}(\mathbf{a}, \mathbf{b})$ is the set of all integer-coordinate points on the straight line connecting nodes $\mathbf{a}$ and $\mathbf{b}$,

$$\mathcal{Q}(\mathbf{a}, \mathbf{b}) = \left\{ \mathbf{p} \in \mathbb{I}_f \mid \frac{\mathbf{p}_y - \mathbf{a}_y}{\mathbf{b}_y - \mathbf{a}_y} = \frac{\mathbf{p}_x - \mathbf{a}_x}{\mathbf{b}_x - \mathbf{a}_x} \text{ and } \mathbf{a}_x \leq \mathbf{p}_x \leq \mathbf{b}_x \right\} \quad (3.4)$$

available in the visible area of the pitch. In the case of players being far apart, we sample a maximum of 30 points on any given line equidistantly. As noted in Equation (3.3), we simply return the full control in instances where the distance between two players is infinitesimally small.

Fraction of personal space control

Similarly to defining the amount of total possible control over a line, we can assign the fraction of all possible control of their surroundings to a given player with the equation

$$PSC(\mathbf{a}, r) = \frac{\sum_{\mathbf{q} \in \mathcal{PS}} PC(\mathbf{q})}{|\mathcal{PS}|}, \quad (3.5)$$

where $\mathcal{PS}$ is the set of all points within a certain radius r of the player node $\mathbf{a}$:

$$\mathcal{PS}(\mathbf{a}, r) = \left\{ \mathbf{p} \in \mathbb{I}_f \mid (\mathbf{p}_x - \mathbf{a}_x)^2 + (\mathbf{p}_y - \mathbf{a}_y)^2 \leq r^2 \right\}. \quad (3.6)$$

For our purposes, we calculate a *small* and a *large* personal space control fraction for $r_{\text{small}} = 3$ and $r_{\text{large}} = 8$, and consider them node features.

Homogeneity of edge control transitions

Recall the definition of the points on a graph edge, corresponding to the direct line between two players from Equation (3.4). In addition to evaluating the fraction of controlled space on this path, we can also evaluate the number of times the majority of pitch control in a given point transitions between teams as we travel along this path. Consider a sequence $\mathcal{S} = (p_1, p_2, \dots, p_n)$, where $p_i \in \mathcal{Q}$ for $i \in 1, 2, \dots, n$ and $p_{x_1} < p_{x_2} < \dots < p_{|\mathcal{Q}|}$. For an arbitrary pitch control function, we can define the indicator of homogeneity $\mathcal{HI}$ for the case that the transition shifts between the two teams more than once across the entire path:

$$\mathcal{HI}(\mathbf{a}, \mathbf{b}) = \begin{cases} 1 & \text{if } \sum_{n=1}^{|\mathcal{S}|} I(s_n, s_{n-1}) > 1 \\ 0 & \text{otherwise,} \end{cases} \quad (3.7)$$

where $I(\mathbf{a}, \mathbf{b})$ is the indicator of a transition in majority pitch control, according to the chosen pitch control function, occurring between two points:

$$I(a, b) = \begin{cases} 1 & \text{if } \left(PC(\mathbf{a}) > 0.5 \right) \neq \left(PC(\mathbf{b}) > 0.5 \right) \\ 0 & \text{otherwise.} \end{cases} \quad (3.8)$$

3.4 Graph Dataset

The final dataset consists of 80,332 graph representations of match formations, recorded at the time of pass events, which ended up being successful. These events were recorded over 166 matches. Figure 3.6 shows the kernel density estimation of the distributions of starting pass locations, as well as their eventual destinations (locations of a successful pass receipt). Notice that the distribution noticeably moves more towards the box in the destination distribution, whereas the modes of both plots remain mostly unchanged; centering at the middle of the pitch, as well as on the attacking wings.

From Figure 3.7, we can see that the vast majority of passes in the dataset are made by backline playmakers, taking the role of center-backs and defensive central midfielders. The distribution of passing angles shows that the majority of passes in the dataset are not progressive, but rather lateral movements of the ball. The pass lengths are expectedly approximately γ -distributed, with the mode of the distribution of pass lengths at ~ 12 yards (11 meters). Figure 3.8 shows that the passes in our dataset generally have a decently sized proportion of all players on the pitch included in the graph(s).

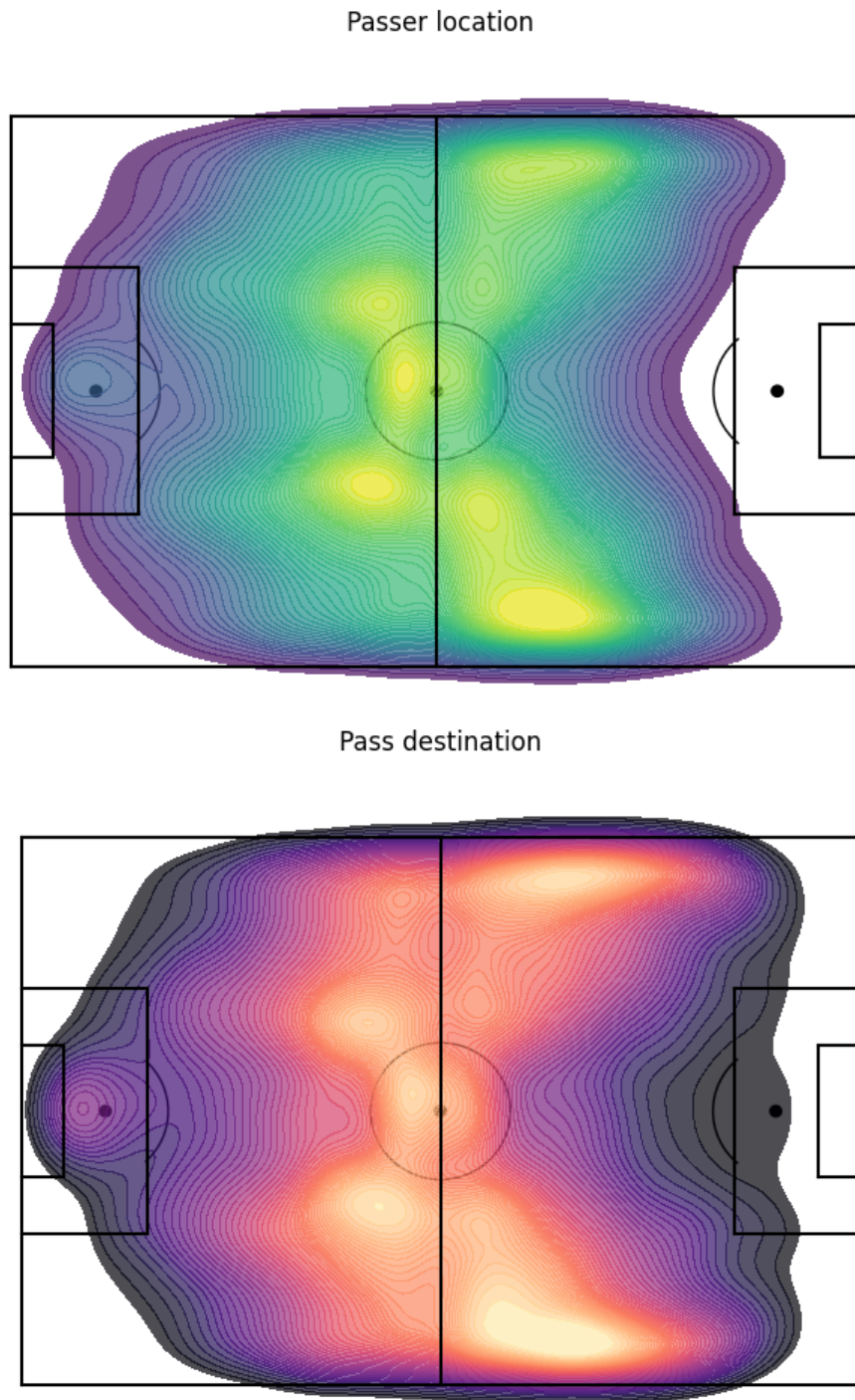


Figure 3.6: Kernel density estimation heatmap of the distribution of passer locations & successful pass destinations across the events, included in our dataset

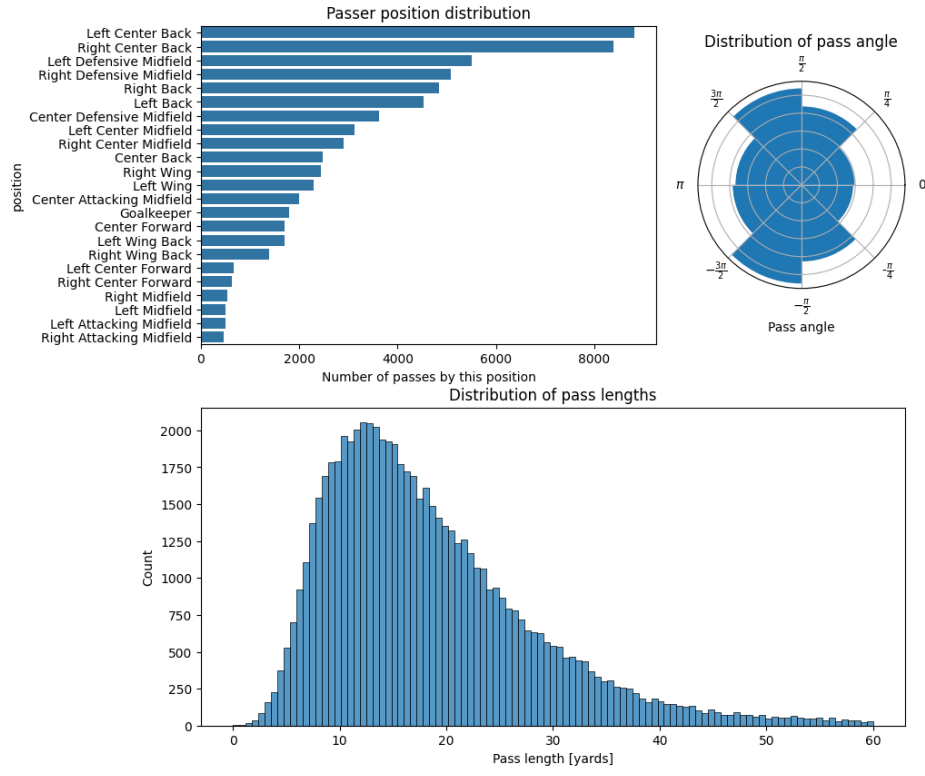


Figure 3.7: General insights into the passes in our dataset.

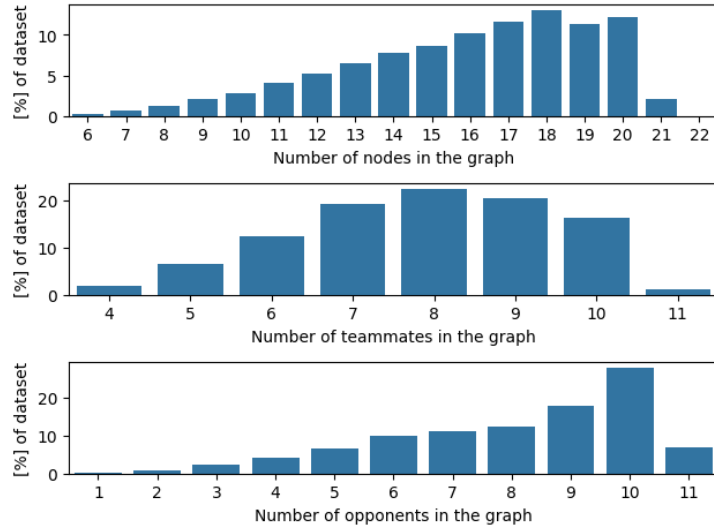


Figure 3.8: Distribution of node counts throughout all the dataset graphs shows that both teams are generally well represented in the data.

3.5 Repository

We make the graph construction implementation available at the GitHub repository <https://github.com/wwidonja/GraphFC>¹. The repository hosts an array of supporting files, tied together by a single master script, located at `graph_construction/create_all_graphs.py`. The script generates the graphs, compatible with both the NetworkX (for SNA and visualization applications), as well as the PyTorch Geometric frameworks (for downstream ML tasks). The final output of the script is an enriched data frame, combining the provider’s datasets – preprocessed, filtered, and transformed as described in the preceding sections, with our final constructed graphs. In the data frame, each row corresponds to one successful pass with three graph variations, made available in each of the two frameworks. Additionally, each row includes an explicit export of the pass’ pitch control data frame (referenced as `control_df` in the source code), including the values of both the Voronoi, as well as the Spearman PC function at every point in the visible area of the pitch. These data frames are used in the computation of PC features, as well as for generating visualizations, such as those in Figure 3.5. In one of the script’s functions, the *include_player_vectors* parameter also allows the user to additionally include our methodology for player identification in the graph construction. When set to *True*, the script additionally labels each node with the hypothesized player names and FIFA vectors. Our methodology for achieving these features is supplementary to this thesis and available in Appendix A. The script supports parallelized graph generation via multiprocessing and is executed in approximately 6 hours on an Apple M3 Max CPU.

For easy dissemination of data about individual graphs in the dataset, we adopt the simple, standardized method of graph summarization from [46] to our domain. An example *network card*, generated by the `generate_network_cards.py` script in our repository, can be seen in Table 3.2.

¹As explicitly sharing and hosting derivative works of the original dataset is not within its terms and conditions, the dataset can not be made available directly.

Name	GraphFC: Match 3788743 — Pass 471
Competition	Europe - UEFA Euro
Match	Belgium - Russia
Competition stage	Group Stage
Match date	2021-06-12
Stadium	Saint-Petersburg Stadium
Timestamp	00:46:50.863
Period	2
Team	Belgium
Passer	Youri Tielemans
Recipient	Toby Alderweireld
# Teammates	10
# Opponents	10
Nodes are	Players of either team
Links are	Abstract interactions between players
Kind	Undirected, Featurized
Onfield Player Centroid	(59.99, 45.56)
Pass Body Part	Right Foot
Pass length	25.62 yards
Pass Height	Ground Pass
Pass End Location	(73.3, 50.50)
Node metadata	Personal space control, Player's (X, Y) coordinates, Team allegiance, Actor & Goalkeeper indicator, Player name, Scaled FIFA vector
Link metadata	Team alignment, direct link pitch control, distance between players, pitch control transition homogeneity
Date of creation	September 2024
Data generating process	Capture of broadcast video footage by StatsBomb via https://github.com/statsbomb/statsbombpy
Rights	University of Ljubljana
Citation	Stropnik (2024)
Access	https://github.com/wwwidonja/GraphFC

Table 3.2: Directly exported LaTeX table for an example pass, as generated by the Network cards script in our repository.

Chapter 4

Modeling Using Our Dataset

We evaluate our dataset by using it to train a diverse array of predictive models, corresponding to different benchmark tasks and graph configurations. The goal of the initial task is to regress the coordinates of a successful pass destination.

We also report on the results of two additional classification tasks – the first being the classification of the receiver’s role in terms of their on-field position. The possible positions (governing the size of the output probability vector) are inferred from the Tactics field of the associated pass, as presented in Table 2.1.

The second classification task is classifying the pass destination into zones of positional play; a domain-specific irregular division of the playing field to some extent implemented by several successful coaches from recent history, such as Johan Cruyff, Hansi Flick, Tomas Tuchel, and Roberto De Zerbi [30]. These coaches view the team as a collection of zones, in relation to which players must adhere in their movements off the ball. It has been well documented that perhaps the most famous proponent of these methods, Pep Guardiola, implements a divided training pitch, as the one seen in Figure 4.1, to drill player behavior in relation to these pitch zones. The positional play model has a middle zone flanked by two half-spaces, with the wing areas divided into even smaller zones. The relatively narrow half-spaces on

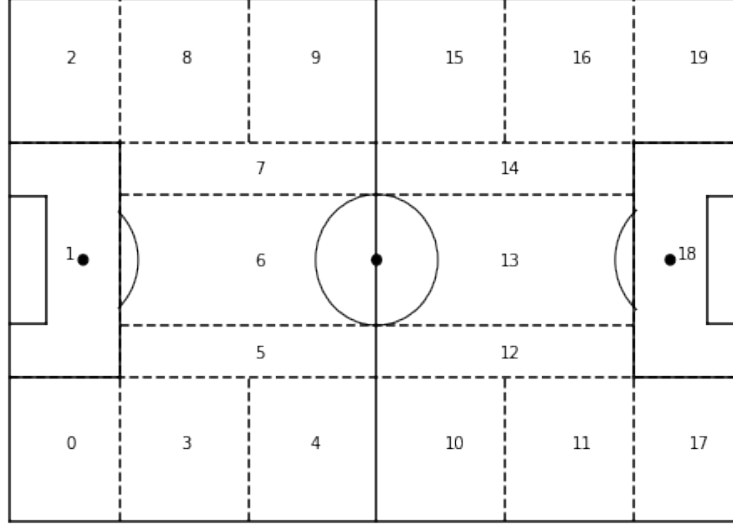


Figure 4.1: The 20 zones of positional play, corresponding to each entry in the target vector in one of the presented classification tasks. The numbers correspond to their indices in the output probability vector.

each side are assigned additional importance in training, citing them as the sources of greatest danger [47]. This problem can be seen as a domain-specific discretization of the pass-destination regression task.

In general, all models use the same set of node and edge features, but a thorough ablation study of these features’ effect on the final performance is summarized in Section 5.1.

In all tasks, our models implement graph deep learning frameworks to generate a latent node feature matrix $\mathbf{H} = f_{\text{GNN}}(\mathbf{X}, \mathbf{A}, \mathbf{E})$. This feature matrix is then further aggregated and transformed to a target dimensionality, appropriate for the relevant task, in one or several prediction heads of any given model.

The models are trained using stochastic gradient descent (SGD) by comparing these final outputs to the observed truths, using mean squared error loss over the Euclidean distances for the regression task, or by minimizing cross-entropy loss for classification.

4.1 Graph Neural Backbone

All of our models share a backbone, employing the sequential operation of combining a node’s feature vector within their first-order neighborhood via message passing. Retaining generality, we can describe a single such operation (hereinafter, a single layer of the GNN) over a node u and its direct first-order neighborhood $\mathcal{N}_u = \{v | (v, u) \in \mathcal{E}\}$, by following the notation from [48] as

$$\mathbf{h}_u^{(l)} = \phi\left(\mathbf{h}_u^{(l-1)}, \bigoplus_{v \in \mathcal{N}_u} \psi(\mathbf{h}_u^{(l-1)}, \mathbf{h}_v^{(l-1)}, \mathbf{e}_{uv})\right), \quad (4.1)$$

where ϕ and ψ can be any learnable function, and $\oplus$ any permutation invariant aggregator. In our instance, we adopt the multi-head Graph Transformer, proposed in [27] as ψ and summation over the neighborhood as $\oplus$. In the transformer, a multi-head attentional mechanism α_{uvi} is deployed over the node representations from the previous layer, and the graph’s edge features, which remain static throughout the network. Equation (4.2) formalizes this definition of ψ and $\oplus$ by factorising it using learnable matrices $\mathbf{W}$. It also shows that the node representations are concatenated across H attention heads and added to the learnable bias;

$$\begin{aligned} \mathbf{h}_u^{(l)} &= \phi\left(\mathbf{W}_0^{(l)} \mathbf{h}_u^{(l-1)} + \left\| \sum_{v \in \mathcal{N}_u} \alpha_{uvi}^{(l)} \left(\mathbf{W}_{2i}^{(l)} \mathbf{h}_v^{(l-1)} + \mathbf{W}_{3i}^{(l)} \mathbf{e}_{uv} \right) \right\| \right), \\ \alpha_{uvi}^{(l)} &= \text{softmax}\left(\frac{(\mathbf{W}_{4i}^{(l)} \mathbf{h}_u^{(l-1)})^\top (\mathbf{W}_{5i}^{(l)} \mathbf{h}_v^{(l-1)} + \mathbf{W}_{6i}^{(l)} \mathbf{e}_{uv})}{\sqrt{d}}\right), \end{aligned} \quad (4.2)$$

where $\mathbf{W}_{1i} \dots \mathbf{W}_{6i}$ are trainable weight matrices for any of the $i \in \{1, 2, \dots, H\}$ attention heads, and d is the hyperparametric size of each head. $\mathbf{W}_0$ corresponds to the weight of the intercept at layer l .

For ϕ , we employ the leaky rectified linear activation function (LeakyReLU) and batch normalization of an otherwise unchanged $\mathbf{h}_u^{(l)}$. In contrast to prior layers, we opt to not concatenate the attention heads in the final layer of our

GNN backbone, but rather average the values over all heads to get the final node feature representations.

4.2 Pooling and Prediction Heads

Given that we are unable to annotate the receiver node from the source freeze frame with absolute certainty, we pose our prediction tasks as graph-level problems and do not predict anything on the node level. To make this possible, an aggregation function $\omega(\mathbf{h}^{(L)}) : \mathbb{R}^{|\mathcal{V}| \times d} \rightarrow \mathbb{R}^{|\mathcal{V}|}$ is required. Our models utilize a simple average over all nodes. Upon training the GNN for any particular downstream task, this pooled representation can also be considered the initial layer of the task prediction head $\mathbf{p}^{(0)}$. In the context of a multi-target model, this representation also corresponds to the shared graph embedding $\mathbf{emb}_{\mathcal{G}}$, retaining general knowledge for all tasks, for which prediction heads are implemented and trained.

$$\mathbf{emb}_{\mathcal{G}} = \mathbf{p}^{(0)} = \omega(\mathbf{h}^{(L)}) = \frac{1}{|\mathcal{V}|} \sum_{v \in \mathcal{V}} \mathbf{h}_v^{(L)} \quad (4.3)$$

Building upon this shared representation, any given prediction head consists of a sequence of fully connected neural network layers, further transforming this hidden representation into the eventual dimensionality of the desired output. This is a 23-dimensional vector when classifying the recipient’s role on the pitch, as 23 is the number of all player positions in the dataset. When regressing the pass position, this is a 2-dimensional coordinate vector. In the case of zone classification, we consider a 20-dimensional vector, corresponding to zonal indices, as shown in Figure 4.1. This transformation is given by the equation

$$\mathbf{p}_n^{(k)} = \text{LeakyReLU}(\text{BatchNorm}(\mathbf{W}_0^{(k)} + \mathbf{W}_n^{(k)} \mathbf{p}^{(k-1)})), \quad (4.4)$$

for every layer of the activation head $k \in 1, 2, \dots, O - 1$, where O is the ordinal number of the output layer of the prediction head, and $\mathbf{W}$ are learnable

trainable weight matrices of the prediction head. In Chapter 5, we report results on models, where $O = 3$.

In the output layer, we do not use the activation functions shown above but rather opt for activations fitting the prediction task. In classification instances, we pass the output vector through a SoftMax activation, normalizing the final activations into a probability vector.

When predicting the player vector, we multiply the prediction with a hyperparametric mask, denoting the player roles, present on the pitch at the time of the pass event. An example of masking the network output for the player position classification task is shown in Equation (4.5):

$$\mathbf{p}_n^{(O)} = M \cdot \text{SoftMax}(\mathbf{W}_0^{(O)} + \sum_{n=1}^{23} \mathbf{W}_n^O \mathbf{p}^{(O-1)}), \quad (4.5)$$

where M is the boolean vector mask, denoting the (up to) 11 positions on the field for the acting team.

In the instance of positional regression, we opt for a sigmoid activation, where we multiply the sigmoid output with $[120, 80]$ to map the activation to the $\mathbb{S}$ space. When training multi-target models, the classification heads all start with the shared layer $\mathbf{emb}_G$, and are unchanged in comparison to their single-target counterparts.

4.3 Training

We train our models using the aforementioned task-appropriate criteria; the mean squared error over the Euclidean distance for regression, and the negative log-likelihood loss to compute the cross entropy between the output probability vector (masked in the case of player classification) and the ground truth one-hot encodings, available for every graph.

In the context of multi-target learning, the joint loss used for back-propagation is a simple summation of these criteria. Our models are trained using an 80/20 percentage split between the training and testing sets. We

optimize the model using the Adam optimization algorithm, which promises computational efficiency with little requirement for hyperparameter fine-tuning [31].

We employ a gradual learning rate decay, triggering every time the loss plateaus over the validation set, calculated after every epoch, for three subsequent epochs. For all learning tasks, we use the starting learning rate of 0.005 and the weight decay coefficient for L2 regularization of 0.00005.

We report results with uncertainty, quantified on 3 cross-validation folds. We use a batch size of $\frac{1}{65}$ of the training dataset. The overall performance estimations, reported in Chapter 5, correspond to the the results of the model iteration that reported the minimal loss on the validation dataset (10 % of the allocated training data, withheld from the training). All results are reported on the held-out test set, never used during training.

Chapter 5

Results

Our best-performing model achieves a median Euclidean distance error of $6.78 \pm .21$ yards ($6.20 \pm .19$ meters). The best-performing classification models achieve a classification accuracy of $55 \% \pm 2 \%$ on the positional play zone classification task and $42 \% \pm 1 \%$ on the player position classification task. Figure 5.1 shows us that the regression error distribution is long-tailed, indicating that the model’s error is even smaller in a large proportion of cases, while still being prone to making extreme mistakes. Figure 5.2 juxtaposes 11 examples of passes, as predicted by our best-performing model, along with their true destinations. Note that, while the examples hint towards a skew in predicting passes further up the field in comparison to their ground truths, this is not a trend that holds over the entire holdout test set.

Figure 5.3 shows results from our zonal prediction task. It indicates that a large fraction of the error comes from the misclassification of passes intended for narrow half-spaces. This is also shown by the large increase in performance when looking at the Accuracy@3 metric – the measure of whether the correct zone is predicted as one of the three zones with the highest probability. There, our best-performing model achieves a score of $85 \% \pm 1 \%$.

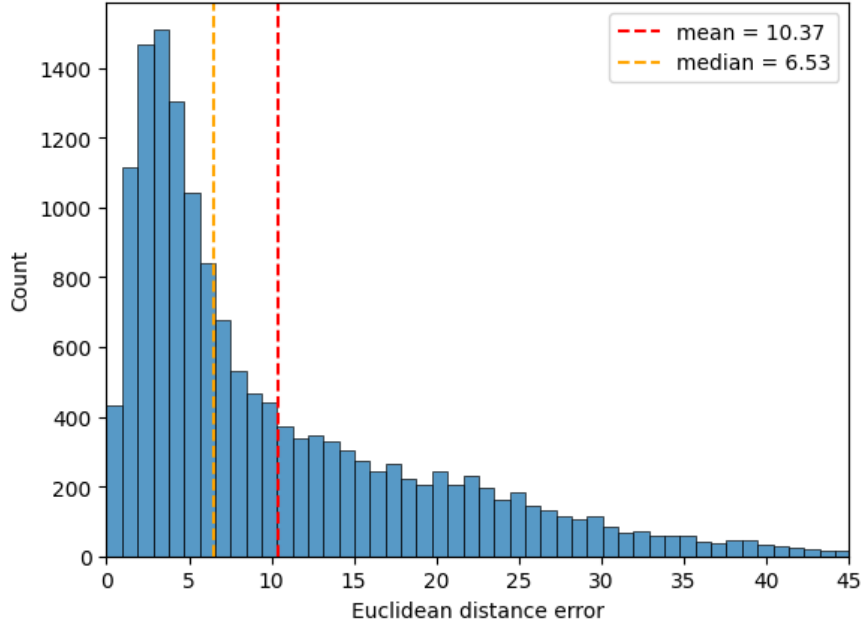


Figure 5.1: Our top performing destination regression model predicts the final pass destination with a median error of ~ 6.53 yards.

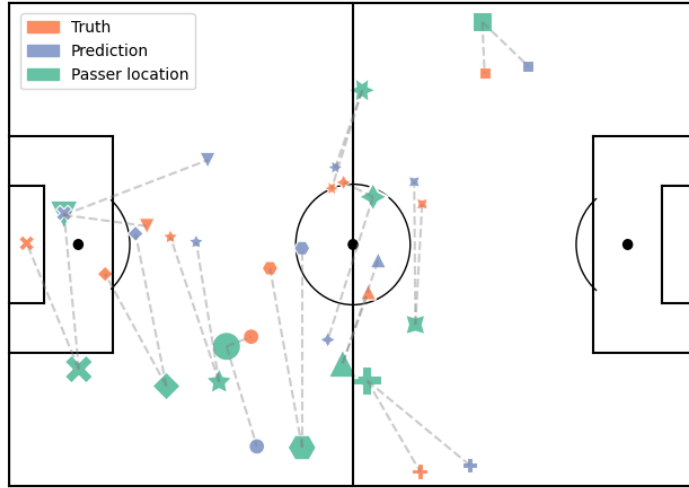


Figure 5.2: 11 examples of predicted pass destinations, along with their corresponding ground-truths, connected to the passer's location. For clarity, each set (passer, truth, prediction) is denoted with a separate set of symbols.

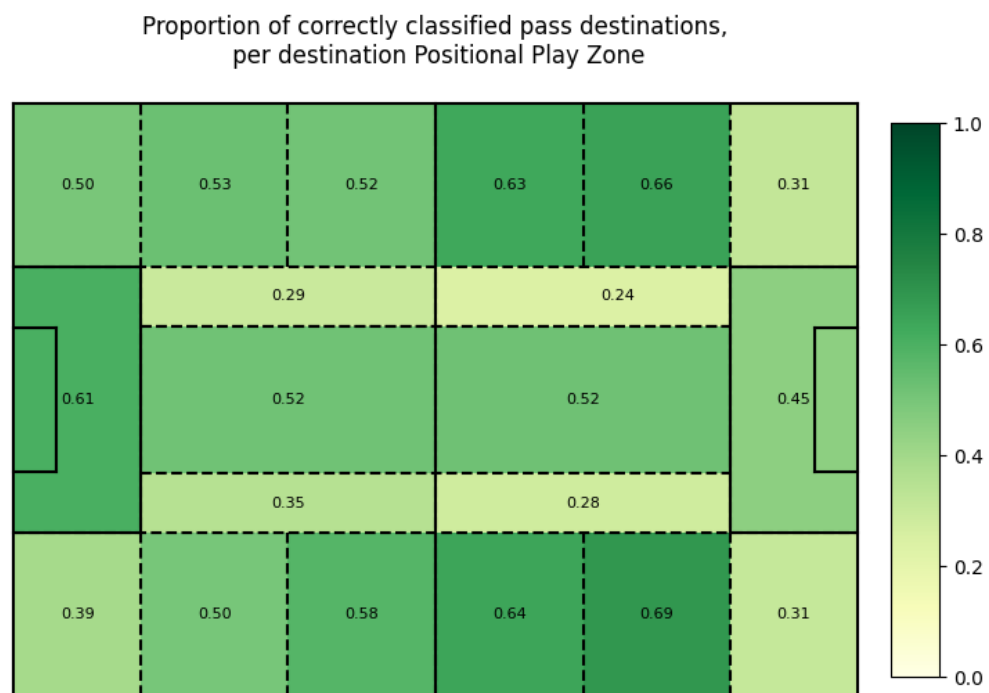


Figure 5.3: Our top-performing pitch zone classification model predicts the correct zone with an accuracy score of 0.54 and an Accuracy@3 score of 0.85. The narrow half-spaces in the middle of the pitch, along with the areas on the far flanks, prove most difficult to predict.

Figure 5.4 shows the confusion matrix for the player-role classification task. Recall that models considered in this context mask the outputs for roles. Consequently, a role, not present on the pitch will never be predicted. Thus, there is little-to-no misclassification between similar roles (i.e. left defensive midfielder (LDM) and left central midfielder (LCM)) – as such roles usually do not coexist on the pitch. Rather, the highest degree of error is observed between players that operate in narrow tandems, i.e. left and right central forwards, and left and right central midfielders. This makes sense, as these roles often tend to rotate to successfully make space to progress the ball. Notice that the error increases to the right side of the plot, where more attacking players are located. This indicates that passes towards attacking players are less predictable. This is also corroborated by the third subplot on the left side of Figure 5.5, where we see that the average error of the regression model increases, when the receiver is in the final quarter of the pitch.

The regression task analysis in Figure 5.5 further shows that the model error increases with the distance of the observed pass, indicating that our model is generally predicting a pass length in the range of the mean of the observed distribution from Figure 3.7. We can notice that passes oriented directly ahead (angle ~ 0) also cause a slight increase in error, as do passes either originating or directed towards the pitch’s Y-coordinate center. This is likely due to a decrease in the ambiguity of the predicted Y component when passing from either flank (i.e. it is more clear at what angle the pass has to be taken from there). The changes in error with respect to the footedness of the pass, as well as the size of the observed graphs, are within the error rate interval and are deemed to not play a big role in the error rate.

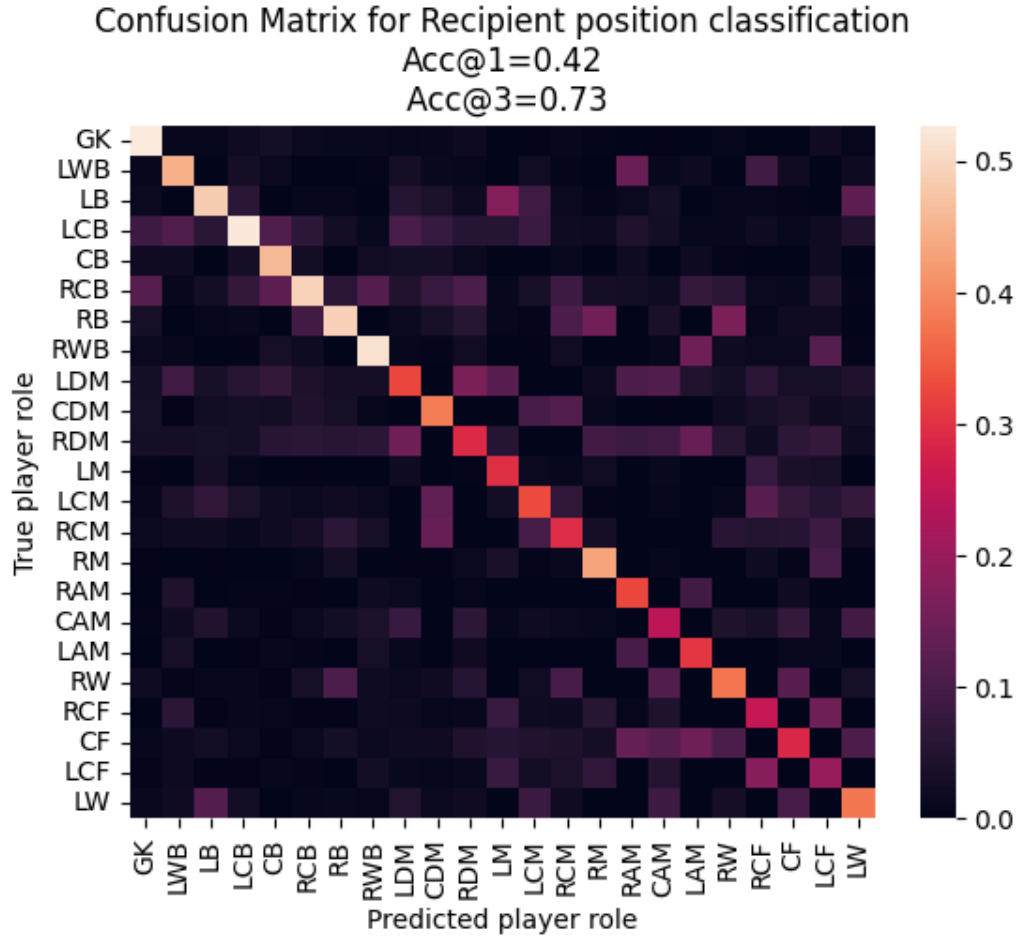


Figure 5.4: Our top-performing receiver role classification model predicts the correct player position with an accuracy score of 0.42. The confusion matrix heatmap shows the frequency of mutual (mis)classification between class labels. Notice how the stability of the prediction decreases when predicting attacking roles on the right side (i.e. RCF, CF, LCF), whereas predicting the backfield players further back (DM, CB) is relatively reliable.

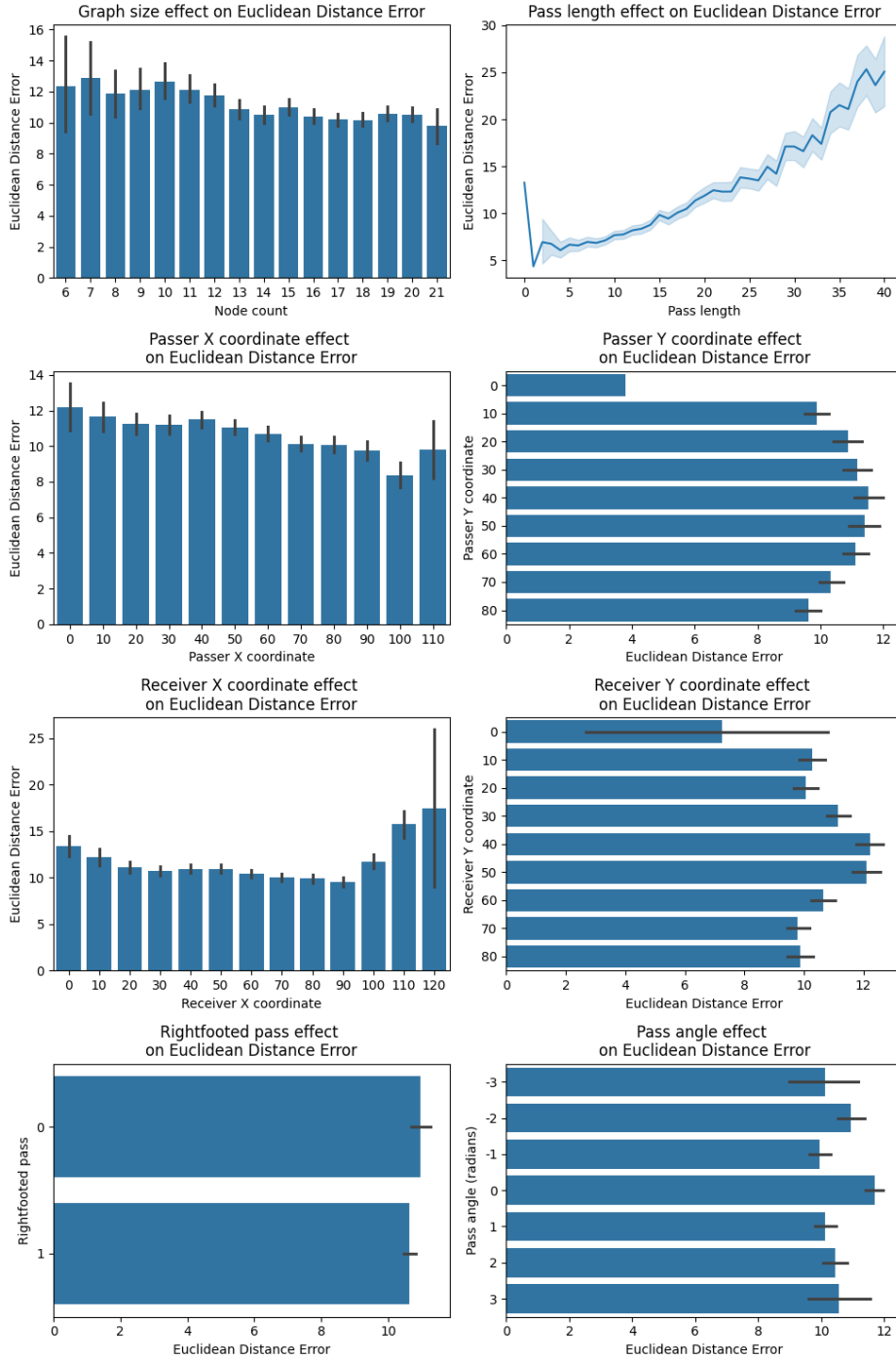


Figure 5.5: Analysis of the effect of different factors on the final error rate, as calculated over a single holdout testing set ($n = 15230$). 95 % confidence intervals of the mean estimations are achieved by bootstrapping the observed distribution.

5.1 Ablation Study

In this section, we report the results of experiments, conducted to understand the effect of different factors on the final predictive performance of our models. These span across different categories; from model architecture, to the differences in graph construction, to the inclusion or omission of specific subsets of information in the training data (graph featurization).

All discussion in this section refers to results, presented in Table 5.2. In the title of each subsection, we also refer to the models relevant to the specific experiment presented in that subsection. The models can be referred to by the same letters in Table 5.2, with the goal of easy referencing between the experiment subsections’ text discussion, and the table. All experiments, reported in the table, are conducted on the regression task, with conclusions presented holding over all other tasks as well, unless stated otherwise.

Does using graphs outperform non-graph methods?

(***K** vs. others*)

We compare the results of our other experiments, utilizing a graph-based backbone, to the Deep Sets model, as presented in [32]. This experiment aims to evaluate the utility of message passing between nodes over (featurised or bare) edges, in comparison to simply aggregating them, while preserving the permutation invariance of the model. The Deep Sets aggregator is given by the function

$$\mathcal{DS}(\mathcal{G}) = \phi \left(\sum_x^{x \in \mathcal{V}} \psi(x) \right), \quad (5.1)$$

where ϕ and ψ are arbitrary multilayer perceptrons. In our implementation, we implement a single-layer encoder and decoder with ReLU activation functions and batch normalization.

Notice, that the model performs worse than some of our graph-based approaches, indicating that message passing does improve performance for the task at hand.

Does using attention-based methods improve performance?

($\mathbf{J}$ vs. $\mathbf{G}$)

We show that attention-based mechanisms outperform a non-attention-based backbone by comparing the Graph Convolutional Network popularized in [12] to a similarly featurized network, utilizing the multi-head attentional backbone of the transformer. It should be noted that the GCN shows exceptionally poor performance, not outperforming even the deep sets aggregation method $\mathbf{K}$, discussed above.

Do trajectory node features improve performance?

($\mathbf{F}$ vs. $\mathbf{D}$)

We train a model, omitting everything but the node features indicating trajectory. We explicitly conduct this experiment using the Voronoi pitch control model, to prevent any information leakage, as the Spearman model uses trajectory information to compute the control values. We can notice that the Euclidean distance error in the implementation, omitting model trajectories (model $\mathbf{F}$) starkly increases when omitting these features, confirming their importance.

Does pitch control improve performance?

($\mathbf{E}$ vs. $\mathbf{A/D}$)

We train a model while omitting pitch control node/edge features. Thus, the graphs are featurized only with player trajectories and trivial edge features, such as the distance between players. We conclude that the improvement in performance for graphs trained with Voronoi pitch-control features is very slight but greater than the margin of error that we attribute to the training split bias.

Does Transformer outperform the SOTA?*(A vs. B)*

The related work [25] found success in analyzing corner kicks by training models, built upon the GATv2 architecture, presented in [24]. We train the model **B** using this architecture and show that the Graph Transformer, presented in our work, shows performance improvements that exceed the uncertainty interval, attributed to training set selection bias.

Does multi-target learning improve performance?*(C vs. D)*

By adding all three of our prediction heads to the model and training it using a composite of all three losses, we hypothesized that the underlying semantics, gathered from reinforcements on all three prediction heads, could lead to a more robust backbone. Notice that model **C** returns more stable and less accurate results, when compared to our best-performing model in Table 5.2. This is the only ablation task where the conclusions between the three tasks disagree, as our model **C** showed the highest accuracy when classifying recipients into zones of positional play. A more thorough comparison of models **C** and **D** between all three tasks can be observed in Table 5.1.

	Regression		Zonal Classification		Recipient Classification	
	Mean	Median	Accuracy	Acc@3	Accuracy	Acc@3
C	10.72 $\pm$.03	07.32 $\pm$.09	0.55 $\pm$.02	0.85 $\pm$.01	0.39 $\pm$.01	0.69 $\pm$.02
D	10.55 $\pm$.16	6.78 $\pm$.21	0.52 $\pm$.01	0.85 $\pm$.02	0.42 $\pm$.01	0.73 $\pm$.01

Table 5.1: Comparison of configurations **C** (Multi-target) and **D** from Table 5.2 across all three tasks. Confidence intervals correspond to the standard deviation of three cross-validation folds. Accuracy@3 (abbreviated as Acc@3) corresponds to the fraction of examples in which the target was within the top 3 most likely recipient candidates.

	Backbone	Trivial EFs	PC EFs	PC NFs	Traj. NFs	Opps?	Multi- Target	Edge spec.	PC Model	Mean Error	Median Error
A	Transformer	✓	✓	✓	✓	✓	✗	<i>FC</i>	<i>Spearman</i>	$10.63 \pm .18$	$07.07 \pm .06$
B	GATv2	✓	✓	✓	✓	✓	✗	<i>FC</i>	<i>Spearman</i>	$10.98 \pm .11$	$7.80 \pm .10$
C	Transformer	✓	✓	✓	✓	✓	✓	<i>FC</i>	<i>Voronoi</i>	$10.72 \pm .03$	$07.32 \pm .09$
D	Transformer	✓	✓	✓	✓	✓	✗	<i>FC</i>	<i>Voronoi</i>	$10.55 \pm .16$	$6.78 \pm .21$
E	Transformer	✓	✗	✗	✓	✓	✗	<i>FC</i>	<i>N/A</i>	$10.80 \pm .09$	$07.08 \pm .06$
F	Transformer	✓	✓	✓	✗	✓	✗	<i>FC</i>	<i>Voronoi</i>	$12.50 \pm .22$	$09.14 \pm .29$
G	Transformer	✓	✓	✓	✓	✗	✗	<i>FC</i>	<i>Voronoi</i>	$11.14 \pm .07$	$7.89 \pm .08$
H	Transformer	✓	✓	✓	✓	✗	✗	<i>HS</i>	<i>Spearman</i>	$11.05 \pm .06$	$7.92 \pm .08$
J	GCN	✗	✗	✓	✓	✓	✗	<i>FC</i>	<i>Voronoi</i>	$17.2 \pm .21$	$15.67 \pm .05$
K	✗ (MLP)	✗	✗	✓	✓	✗	✗	<i>N/A</i>	<i>Spearman</i>	$13.60 \pm .07$	$12.04 \pm .04$

Table 5.2: Results on the pass destination regression coordinate task for the following models: **A** – Spearman Pitch Control model with Transformer Backbone, **B** – Spearman Pitch Control model with GATv2 Backbone, **C** – Voronoi pitch control model (same as D), but trained with multiple prediction heads, optimized with a composite loss, **D** Transformer with Voronoi pitch control model, **E** Model without any pitch-control-based features, **F** Model without (implicit or explicit) information about players’ movement trajectories, **G** Model with no opposition nodes included, **H** Transformer-based model on the hub-and-spokes configuration, **J** Same as D with non-attentional backbone without edge features, **K** Baseline without explicit usage of graph structure (simple aggregation). All confidence intervals correspond to the standard deviation of 3 cross-validation folds.

Chapter 6

Conclusion

In this work, we first introduced one of the more expansive datasets of football match events, available openly to the general public. We thoroughly introduced methodologies for overcoming some problems that arose when constructing useful graph representations of match moments, described in the dataset. One of these problems was the choice of the graphs’ neighborhood structures – for which we show that a fully connected network, including nodes of both teams, offers the most information. Another methodology, heavily discussed in our work, is the computation of player trajectory vectors, encoded as node features in our graphs. We present the methodology to compute them and show experimental results, confirming the utility of their inclusion in downstream ML tasks. This thesis also overviews the notion of pitch control, useful for network visualization, and compares different modeling methods to achieve the best predictive results. Our results demonstrate that our approach effectively models the passer’s decision-making and that graph models enriched with appropriate features can significantly contribute to advanced football analytics.

We leave several improvements upon the proposed methodology as subjects for future work. First, the *Statsbombpy* package offers a plethora of resources, required to prepare a similar dataset for the domain of women’s football – initially discarded from this research due to the lack of quality

player data, required to annotate the nodes – a separate effort of the early stages of our work, presented in Appendix A. Alternatively, building a similar dataset upon resources that do not suffer from the key limitations of our source (i.e. player anonymity) would likely boost performance – as we hypothesize that player identity could play a factor in the actor’s pass choice, and more accurate trajectory vectors could be calculated. Alternate areas that might yield performance improvements are the addition of global match-level features, modeling different lengths of passes independently, modeling the stochasticity of the passer’s decision via Bayesian methods, or otherwise estimating the destination’s probability distribution instead of an absolute coordinate. Alternate data sources, that might allow for modeling player fatigue by tracking their game involvement throughout the observed match, could also prove interesting for future work in the field.

We make the graph construction implementation available at the GitHub repository github.com/wwidonja/GraphFC. The script, presented therein, generates the graphs, compatible with both the NetworkX, as well as the PyTorch Geometric framework. The repository also hosts a script for generating succinct summaries of individual graphs, appropriate for dissemination to the SNA community.

Appendix A

Appendix: Player Identification

A.1 Player Data

Table 2.6 shows the attributes from the FIFA dataset introduced in Section 2.3 that we utilize for matching to the provider’s event data from Tables 2.1 and 2.5, or as features of our graph nodes. We supplement table 2.6 here with Table A.1. The features described in it exhibit a high degree of correlation and are often summarized by a varying number of macro attributes in the associated video games. As these groupings vary between game iterations or game modes and do not hold the same across all players (i.e. goalkeepers are summarized by a different set of attributes, but are featurized similarly in our graphs), we opt to summarize them ourselves by calculating their six principal components.¹

We do this by first standardizing the data to zero mean and unit variance, then computing the sets’ covariance matrix, and finally performing its eigenvalue decomposition to find the 6 linearly uncorrelated (eigen)vectors with maximum variance [49]. The choice of the number of principal components here corresponds to the results of a Scree test, visualized in Figure A.1. Note that the number of semantic *macro-categories*, traditionally included in the

¹Opting for PCA-reduced features or a subset of the features from the dataset did not significantly impact our results in early experimentation.

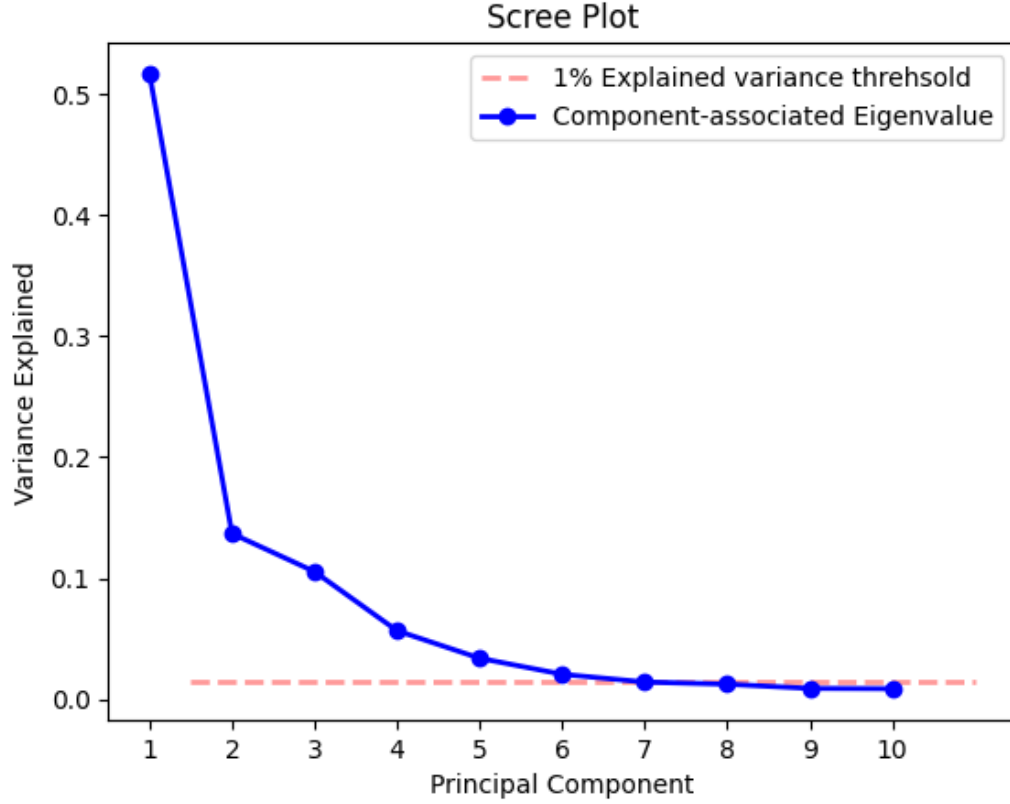


Figure A.1: Scree plot for principal component analysis of FIFA data from Tables 2.6 and A.1

FIFA series of games, is 6-8, depending on the game iteration.

We scale each feature to the 0-1 interval, considering all players available in a given game, and cast the players' footedness as a one-hot encoded variable, leaving us with a 7-dimensional feature vector, describing an individual players' on-field abilities, general renown, and playing preferences.

A.2 Player Information on Nodes

As discussed in Section 3.2, the lack of identification of players in the Freeze frame dataset prevents us from easily annotating our graphs's nodes with the player information from the video game dataset. To achieve an as-reliable-

Category	Attributes
Attacking	Crossing, finishing, heading, short passing, attacking volleys.
Skill	Curve, fk accuracy, long passing, ball control.
Movement	Acceleration, sprint speed, agility, reactions, balance.
Power	Shot power, jumping, stamina, strength, long shots.
Mentality	Aggression, interceptions, positioning, vision, penalties, composure.
Defending	Marking awareness, standing tackle, sliding tackle.
Goalkeeping	Diving, handling, kicking, positioning, reflexes.

Table A.1: Additional FIFA attributes, implicitly used as node features in our player-annotated dataset (as summarized by their joint principal components). The attributes are grouped by their common category, as presented in the source data tables.

as-possible approximation of player identity, we, again, posit this as an assignment problem. From the *Tactics* attribute of the most recent *Starting XI* or *Tactical Shift* events from Table 2.2, we are able to see the names and positions of all players, present on the pitch on any given time. By observing the distribution of all on-ball events in the match for any given player, we can estimate the probability density function (PDF) of the player being located in any given area of the pitch. We can similarly estimate the movement map of a given player position, but, rather than referring to the specific match, we can estimate the PDF over all of the matches that had a player in that position. We perform these estimates using a Gaussian kernel density estimation, formalized as follows:

Let $(x_1, x_2, \dots, x_n)$ be independent and identically distributed samples from a univariate distribution with an unknown density f at point x . We define $\hat{f}_h$ as its Kernel density estimator

$$\hat{f}_h(x) = \frac{1}{nh} \sum_{i=1}^n K\left(\frac{x - x_i}{h}\right), \quad (\text{A.1})$$

where K is a non-negative Kernel function and h is the smoothing bandwidth parameter. For our crude estimate, we settle for a unit bandwidth and a Gaussian kernel – a standard normal distribution with a null mean and unit variance. Explicitly stating the above for a two-dimensional point p with coordinates p_x and p_y , and using the definition of the standard normal distribution, we get the formulation for our kernel density estimation in Equation (A.2);

$$\hat{f}(p) = \frac{1}{n} \sum_{i=1}^n \frac{1}{2\pi} \exp\left\{-\frac{1}{2}[(p_x - x_i)^2 - 2(p_x - x_i)(p_y - y_i) + (p_y - y_i)^2]\right\}. \quad (\text{A.2})$$

With kernel density estimations for all on-field players in the match, as well as their positions across the dataset, we can evaluate the likelihood of any given player being located at any of the observed points in two ways. A graphical representation of this is shown in Figure A.2. By averaging² the scaled probability densities, and taking their negative value, we can compute a cost for each player being located at the given point at the time of sampling. From here, we employ the same steps as presented in Section 3.2, to solve this assignment problem using the Hungarian Algorithm.

Like before, we solve the assignment problem separately for the in-possession and out-of-possession team, using the inverted pitch coordinates for computing the opposition KDE. We also exclude goalkeepers and passers from the assignment, as we can directly infer these player’s identities from the event data. We assign each node their corresponding player’s 7-dimensional vector, as well as the one-hot encoding of the position they are referenced as occupying in the associated *Tactics* event. An example mapping of players in the possession team is shown in Figure A.3. Domain experts will observe that, while a plausible mapping is presented in the figure, the defensive midfielders positioning off-the-ball might be slightly too deep. Such inconsistencies stem

²We observed that a slight weighing in favor of the player’s observed match movement produced better results, so we use a 60 % / 40 % weighted sum in our final implementation. This averaging had a strong impact on the final assignment in any given frame.

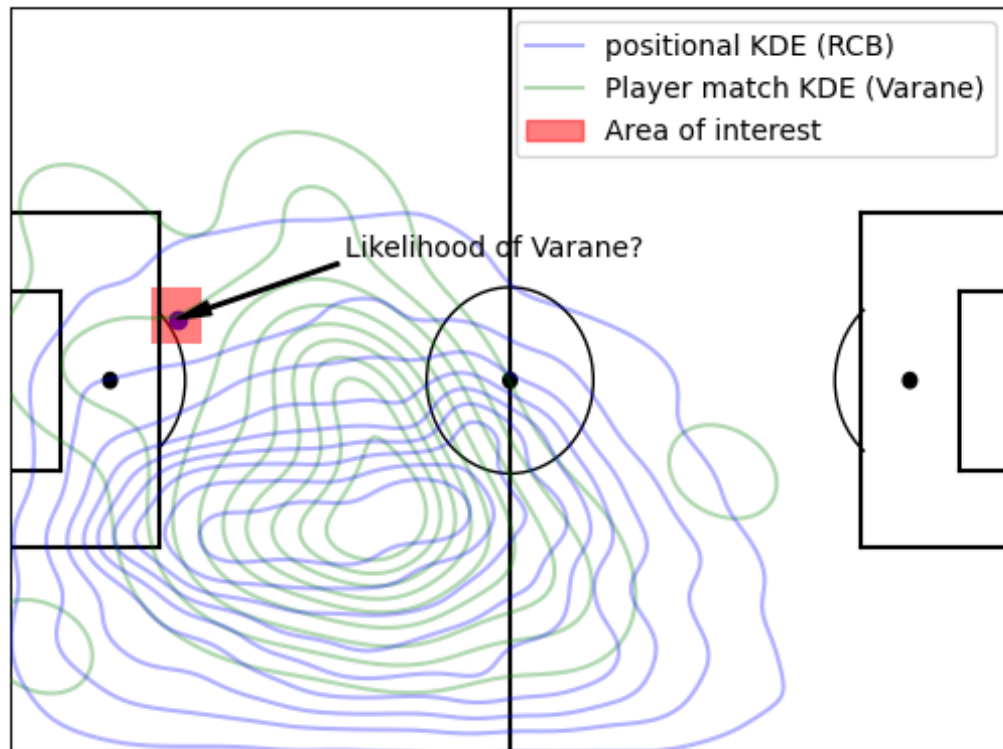


Figure A.2: The KDE plots of a particular player (Raphaël Varane) and their position (RCB), overlaid over an anonymous point. By integrating the two PDF estimates over each point's surroundings, we generate the cost matrix, used in the assignment of players to all observed points.

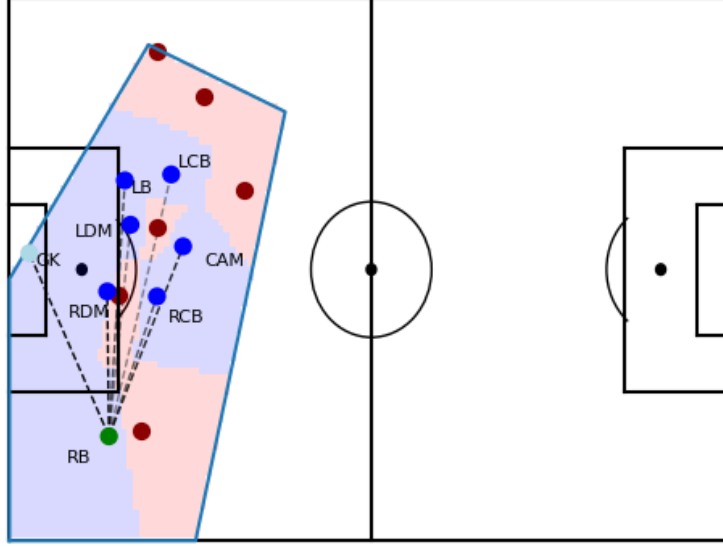


Figure A.3: An example player mapping on the French in-possession team, using the approach from Section A.2, annotated with the player’s positions.

from our ground-truth events (from which the kernel density estimates are calculated) corresponding only to on-ball events, whereas the team’s off-ball shape is fundamentally different. We attribute this limitation of our data sources as one of the reasons our downstream predictive results are not improved by adding the players’ feature vectors to the considered node features, as we report in the following section. Consequently, we treat these features as approximations, as the cost of the evaluation of their accuracy, on the presented data, would be prohibitively time-consuming and is out of the scope of the presented work.

A.3 Experimental Results

We consider the best-performing combination of the GNN architecture, graph structure, and featurization specification (i.e. example **D** from Table 5.2) as the base for our experimentation. We train the same network on the same

	Euclidean Distance Error			
	Mean		Median	
	no PVs	PVs	no PVs	PVs
Edge Conf.				
FC & Opps	10.55 $\pm$.16	10.73 $\pm$.12	6.78 $\pm$.21	7.45 $\pm$.14
HS	11.14 $\pm$.07	10.91 $\pm$.10	7.89 $\pm$.09	7.91 $\pm$.11

Table A.2: Comparison of results for the task from 5.2 with added player vectors. Confidence intervals correspond to the standard deviation of from three cross-validation folds.

three cross-validation splits and allow it to train for the same maximum amount of epochs. Table A.2 shows the results for this experiment, working on the less-favorable *hub and spokes* graph configuration. Notice that the addition of the player feature vectors did not improve test results on the best-performing edge configuration and that, while a performance increase was notable for the Hub and Spokes configuration, it is very slight and within the standard deviation of the cross-validation variation in its median estimate.

Bibliography

- [1] S. Schmidt, The importance of friendships for academic success, *Journal of Food Science Education* 19 (2020) 2–5.
- [2] K. Tuyls, S. Omidshafiei, P. Muller, Z. Wang, J. Connor, D. Hennes, I. Graham, W. Spearman, T. Waskett, D. Steel, et al., Game plan: What ai can do for football, and what football can do for ai, *Journal of Artificial Intelligence Research* 71 (2021) 41–88.
- [3] 2023 sports fan insights: The beginning of the immersive sports era — www2.deloitte.com, [Accessed 2024-08-25].
URL `https://www2.deloitte.com/us/en/insights/industry/media-and-entertainment/immersive-sports-fandom.html`
- [4] I. B. da Costa, L. B. Marinho, C. E. S. Pires, Forecasting football results and exploiting betting markets: The case of “both teams to score”, *International Journal of Forecasting* 38 (3) (2022) 895–909.
- [5] R. Goka, Y. Moroto, K. Maeda, T. Ogawa, M. Haseyama, Prediction of shooting events in soccer videos using complete bipartite graphs and players’ spatial-temporal relations, *Sensors* 23 (9) (2023) 4506.
- [6] S. Omidshafiei, D. Hennes, M. Garnelo, E. Tarassov, Z. Wang, R. Elie, J. T. Connor, P. Muller, I. Graham, W. Spearman, et al., Time-series imputation of temporally-occluded multiagent trajectories (2021). `arXiv:2106.04219`.

-
- [7] S. Omidshafiei, D. Hennes, M. Garnelo, Z. Wang, A. Recasens, E. Tarassov, Y. Yang, R. Elie, J. T. Connor, P. Muller, et al., Multia-gent off-screen behavior prediction in football, *Scientific reports* 12 (1) (2022) 8638.
 - [8] G. Kumar, *Machine learning for soccer analytics*, University of Leuven (2013).
 - [9] D. Araújo, K. Davids, Team synergies in sport: theory and measures, *Frontiers in psychology* 7 (2016) 1449.
 - [10] P. Veličković, G. Cucurull, A. Casanova, A. Romero, P. Liò, Y. Bengio, Graph attention networks, in: *International Conference on Learning Representations*, 2018.
 - [11] W. L. Hamilton, R. Ying, J. Leskovec, Inductive representation learning on large graphs, in: *Proceedings of the 31st International Conference on Neural Information Processing Systems*, 2017, pp. 1025–1035.
 - [12] T. N. Kipf, M. Welling, Semi-supervised classification with graph convolutional networks, in: *International Conference on Learning Representations*, 2017.
 - [13] K. Xu, W. Hu, J. Leskovec, S. Jegelka, How powerful are graph neural networks?, in: *International Conference on Learning Representations*, 2019.
 - [14] T. U. Grund, Network structure and team performance: The case of english premier league soccer teams, *Social Networks* 34 (4) (2012) 682–690.
 - [15] F. M. Clemente, F. M. L. Martins, D. Kalamaras, P. D. Wong, R. S. Mendes, General network analysis of national soccer teams in fifa world cup 2014, *International Journal of Performance Analysis in Sport* 15 (1) (2015) 80–96.

-
- [16] S. McLean, P. M. Salmon, A. D. Gorman, M. Naughton, C. Solomon, Do inter-continental playing styles exist? using social network analysis to compare goals from the 2016 euro and copa football tournaments knock-out stages, *Theoretical Issues in Ergonomics Science* 18 (4) (2017) 370–383.
 - [17] J. Castellano, I. Echeazarra, Network-based centrality measures and physical demands in football regarding player position: Is there a connection? a preliminary study, *Journal of sports sciences* 37 (23) (2019) 2631–2638.
 - [18] F. Korte, D. Link, J. Groll, M. Lames, Play-by-play network analysis in football, *Frontiers in psychology* 10 (2019) 1738.
 - [19] A. Rathke, An examination of expected goals and shot efficiency in soccer, *Journal of Human Sport and Exercise* 12 (2) (2017) 514–529.
 - [20] T. Decroos, L. Bransen, J. Van Haaren, J. Davis, Actions speak louder than goals: Valuing player actions in soccer, in: *Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining*, 2019, pp. 1851–1861.
 - [21] C. Merhej, R. Beal, S. Ramchurn, T. Matthews, What happened next? using deep learning to value defensive actions in football event-data (2021). [arXiv:2106.01786](https://arxiv.org/abs/2106.01786).
 - [22] M. Stöckl, T. Seidl, D. Marley, P. Power, Making offensive play predictable-using a graph convolutional network to understand defensive performance in soccer, in: *Proceedings of the 15th MIT Sloan Sports Analytics Conference*, Vol. 2022.
 - [23] J. C. Campos, Determining the phases of play using graph neural network embeddings, in: *Proceedings of the annual StatsBomb Conference 2021*, Vol. 2021.

-
- [24] S. Brody, U. Alon, E. Yahav, How attentive are graph attention networks?, in: International Conference on Learning Representations, 2022.
- [25] Z. Wang, P. Veličković, D. Hennes, N. Tomašev, L. Prince, M. Kaisers, Y. Bachrach, R. Elie, L. K. Wenliang, F. Piccinini, et al., Tacticalai: an ai assistant for football tactics, *Nature communications* 15 (1) (2024) 1906.
- [26] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. Kaiser, I. Polosukhin, Attention is all you need, in: Proceedings of the 31st International Conference on Neural Information Processing Systems, NIPS’17, Curran Associates Inc., Red Hook, NY, USA, 2017, p. 6000–6010.
- [27] Y. Shi, Z. Huang, S. Feng, H. Zhong, W. Wang, Y. Sun, Masked label prediction: Unified message passing model for semi-supervised classification, in: Z.-H. Zhou (Ed.), Proceedings of the Thirtieth International Joint Conference on Artificial Intelligence, IJCAI-21, International Joint Conferences on Artificial Intelligence Organization, 2021, pp. 1548–1554, main Track.
- [28] munkres for Python, The hungarian algorithm implementation in python, [Accessed 2024-09-01].
URL <http://software.clapper.org/munkres/>
- [29] W. Spearman, Quantifying pitch control, in: OptaPro Analytics Forum, 2016.
- [30] TIFO Football, What is positional play?, [Accessed 2024-09-01].
URL <https://www.youtube.com/watch?v=dhmCwI5vA1E>
- [31] D. Kingma, J. Ba, Adam: A method for stochastic optimization, in: International Conference on Learning Representations (ICLR), San Diego, CA, USA, 2015.

-
- [32] D. Buterez, J. P. Janet, S. J. Kiddle, D. Oglic, P. Liò, Graph neural networks with adaptive readouts (2022). [arXiv:2211.04952](#).
- [33] M.-C. Zheng, L.-J. Wang, Enhancing low basketball experience viewer broadcast experience via data visualization: The national basketball association case study, *Human Factors and Systems Interaction* (2024) 310.
- [34] V. G. Satorras, E. Hoogeboom, M. Welling, E (n) equivariant graph neural networks, in: *International conference on machine learning*, PMLR, 2021, pp. 9323–9332.
- [35] M. Marchiori, M. de Vecchi, Secrets of soccer: Neural network flows and game performance, *Computers Electrical Engineering* 81 (2020) 106505, [Accessed 2024-09-01].
URL <https://www.sciencedirect.com/science/article/pii/S0045790618332518>
- [36] J. Devlin, Bert: Pre-training of deep bidirectional transformers for language understanding (2018). [arXiv:1810.04805](#).
- [37] D. Ding, H. H. Huang, A graph attention based approach for trajectory prediction in multi-agent sports games (2020). [arXiv:2012.10531](#).
- [38] C. Pfaff, E. Hunter, H. Hong, D. Forestell, A. Fialkov, Z. Drassinower, T. Chan, Controlling ball progression in soccer (2022). [arXiv:2210.16474](#).
- [39] StatsBomb, Open data, [Accessed 01-09-2024] (2018).
URL <https://github.com/statsbomb/open-data>
- [40] StatsBomb, Statsbombpy, [Accessed 2024-09-01] (2020).
URL <https://github.com/statsbomb/open-data>
- [41] Sofifa, [Accessed 2024-09-01].
URL <https://sofifa.com/>

-
- [42] S. Leone, Kaggle: Fifa dataset compilation, [Accessed 2024-09-01].
URL <https://www.kaggle.com/datasets/stefanoleone992/fifa-22-complete-player-dataset>
- [43] S. Leone, Kaggle: Ea fc 24 dataset, [Accessed 2024-09-01].
URL <https://www.kaggle.com/datasets/stefanoleone992/ea-sports-fc-24-complete-player-dataset>
- [44] J. Munkres, Algorithms for the assignment and transportation problems, *Journal of the Society for Industrial and Applied Mathematics* 5 (1) (1957) 32–38.
- [45] W. Spearman, A. Basye, G. Dick, R. Hotovy, P. Pop, Physics-based modeling of pass probabilities in soccer, in: *Proceeding of the 11th MIT Sloan Sports Analytics Conference*, Vol. 1, 2017.
- [46] J. Bagrow, Y.-Y. Ahn, *Network cards: concise, readable summaries of network data* (2022).
- [47] Breaking The Lines, Tactical analysis: What is juego de posición?, [Accessed 2024-09-01].
URL <https://breakingthelines.com/tactical-analysis/what-is-juego-de-posicion/>
- [48] M. M. Bronstein, J. Bruna, T. Cohen, P. Veličković, Geometric deep learning: Grids, groups, graphs, geodesics, and gauges (2021). *arXiv*: 2104.13478.
- [49] I. T. Jolliffe, J. Cadima, Principal component analysis: a review and recent developments, *Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences* 374 (2065) (2016) 20150202.